

GOBI-638455



Batch_____

EXTERNAL EXAMINER

CCS352 MULTIMEDIA AND ANIMATION

OBJECTIVES:

- To grasp the fundamental knowledge of Multimedia elements and systems
- To get familiar with Multimedia file formats and standards
- To learn the process of Authoring multimedia presentations
- To learn the techniques of animation in 2D and 3D and for the mobile UI
- To explore different popular applications of multimedia

LIST OF PROGRAMS:

1. Working with Image Editing tools
2. Working with Audio Editing tools
3. Working with Video Editing and conversion tools
4. Working with web/mobile authoring tools
5. Working with Animation tools
6. Working with E-Learning authoring tools
7. Working with VR And AR Applications.

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4		Working with web/mobile authoring tools			
5		Working with Animation tools			
6		Working with E-Learning authoring tools			
7		Working with VR And AR Applications.			

EX.NO : 1	WORKING WITH IMAGE EDITING TOOLS
DATE :	

AIM:

To Use different selection and transform tools to modify or improve an image using Image Editing Tools.

ALGORITHM:

Step 1 : Open the web browser and install the krita application on your system.

Step 2 : open krita application and perform selection and transform operation to modify an image.

Step 3 : First create the new blank page and select the image.

Step 4 : Move the tool option to select the preview and choose the fast option.

Step 5 : Select the options for position ,rotate, scale and shear to increase or reduce the image size.

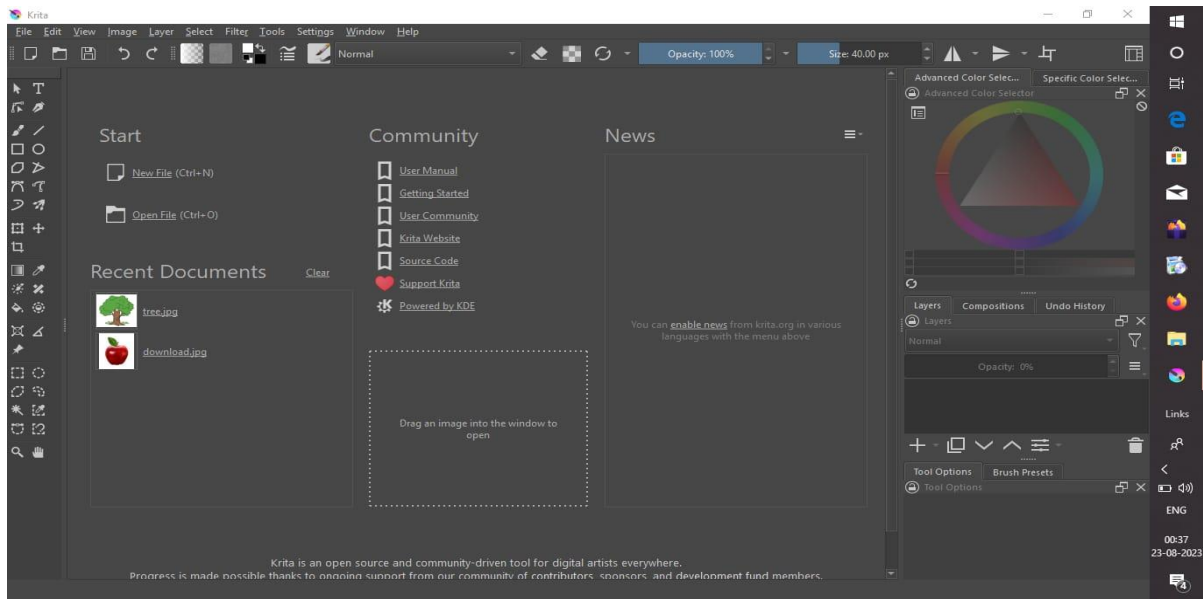
Step 6: Select to tool options for free, precptive, warp ,cage, liquefy and mesh are used for transforms to the image

Step 7: Finally you get the output

PROCEDURE :

Step1:

Select for the new blank page (ctrl +n)

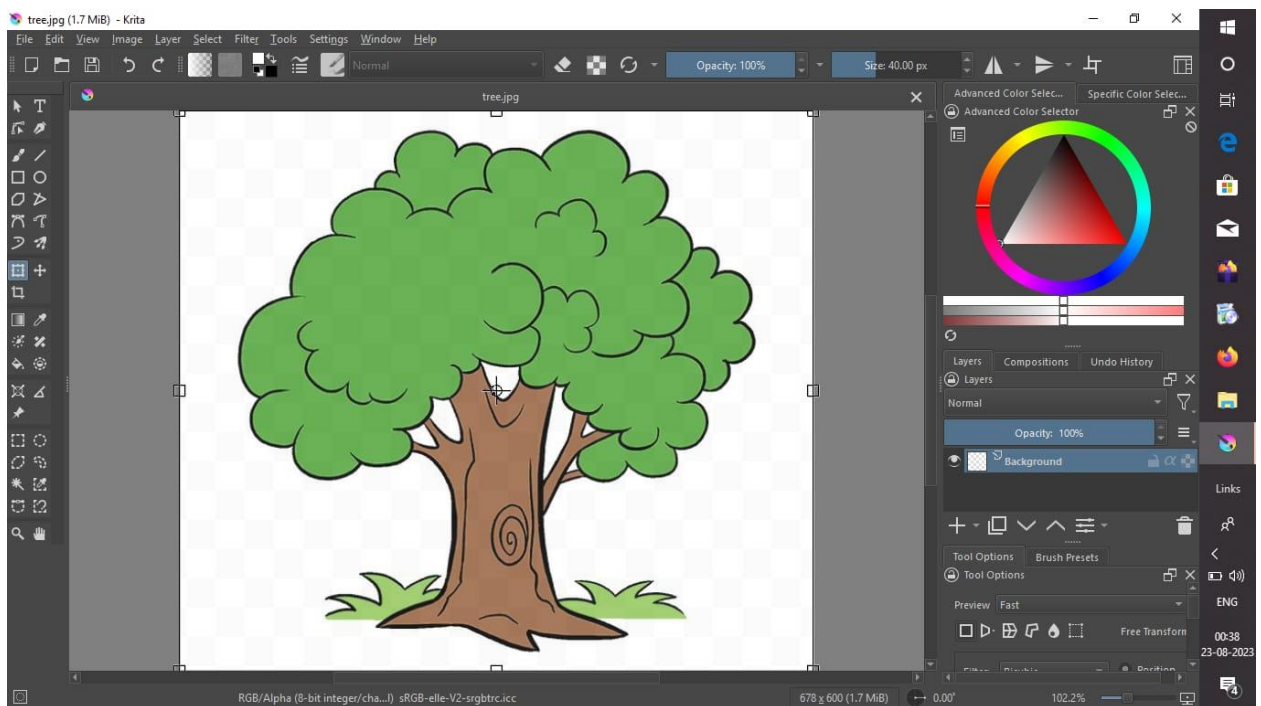


Step 2:

Import the image.

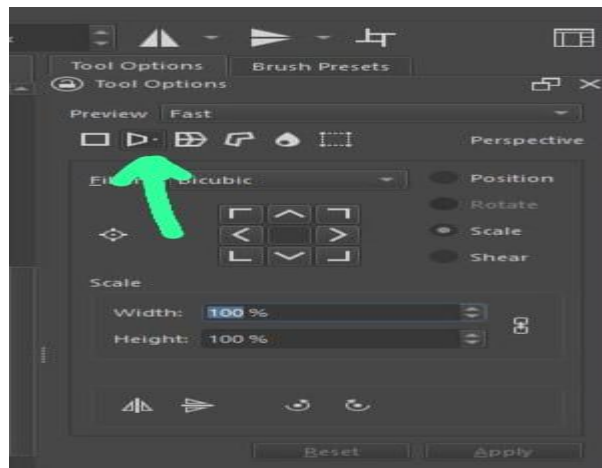
select to the fast option.

The fast option used to reduce lag and speed the process.



Step 3:

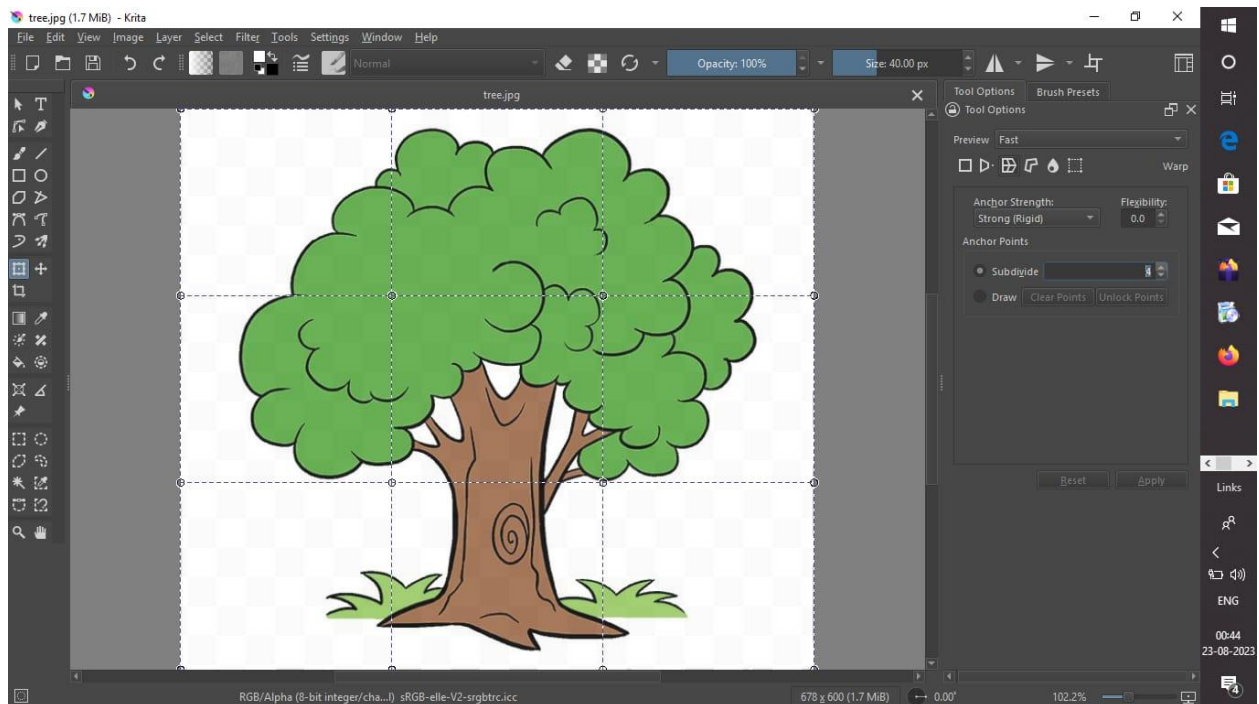
By choosing the perspective option to increase or reduce the width and length of the image.



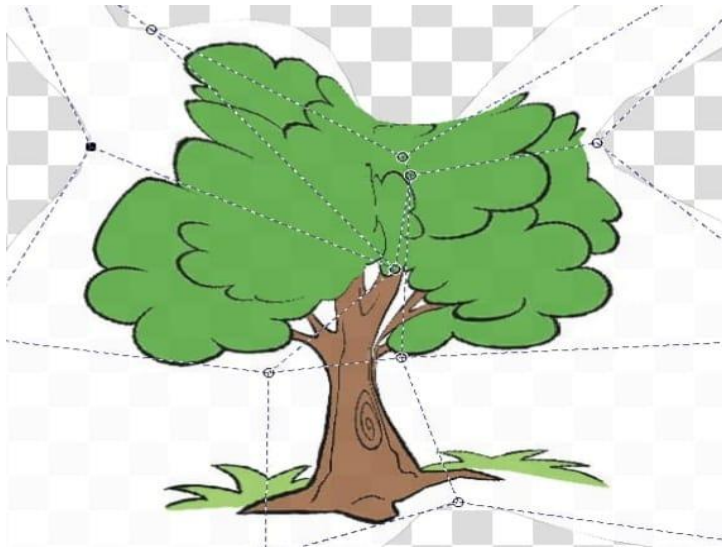
Step 4:

Next select warp option to apply transparent effect to an image surface

Before image:



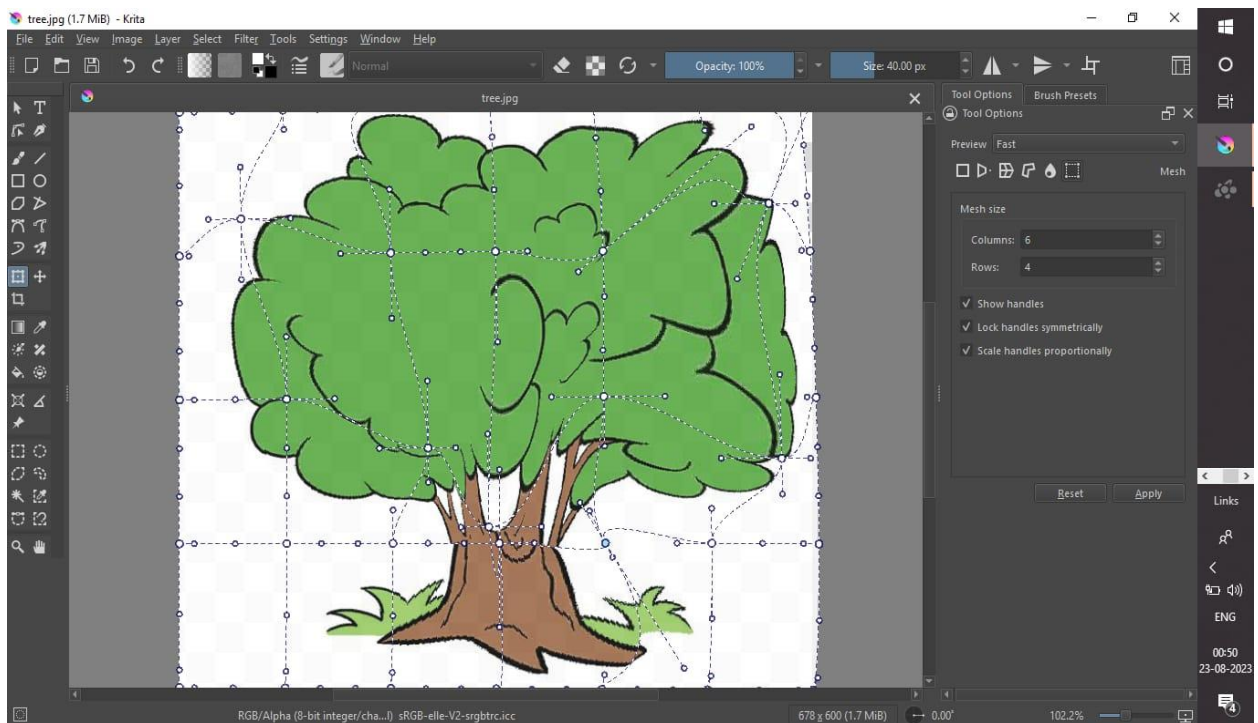
After transparent image:

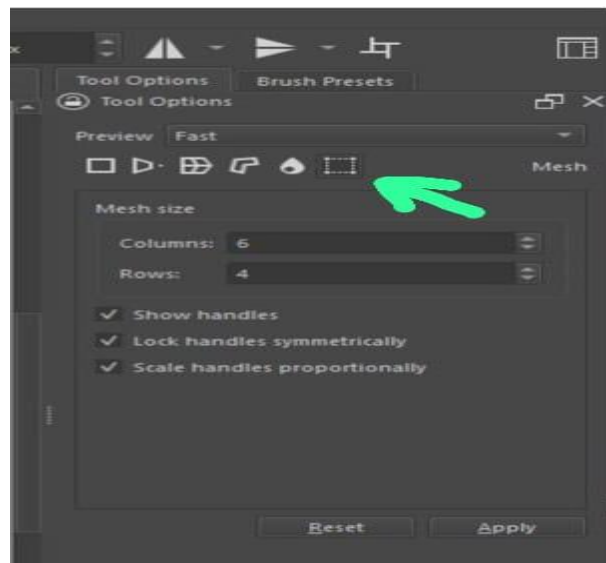


Step 5:

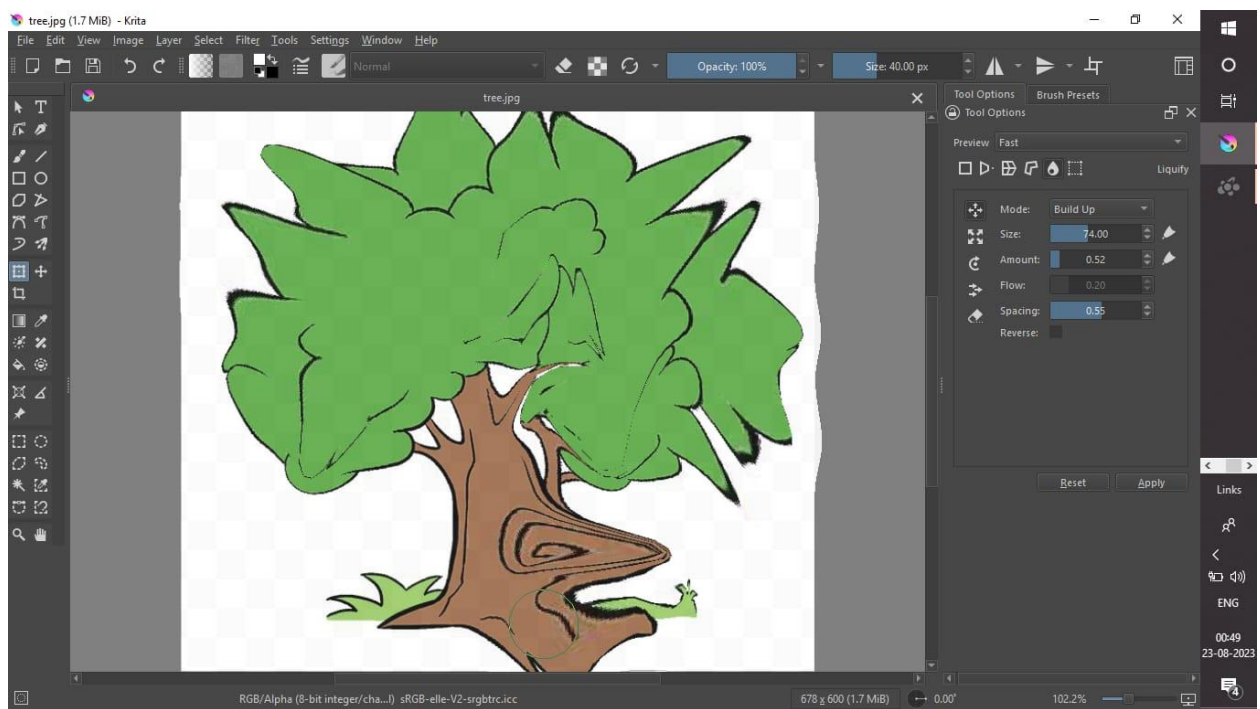
After applying the transparent effect you to fix the columns and rows by using mesh option.

Before image:





After image:



RESULT:

Thus the selection and transform tools to modify or improve an image using Image Editing Tools was implemented and verified successfully.

EX.NO : 2	WORKING WITH AUDIO EDITING TOOLS
DATE :	

AIM:

To perform audio compression by choosing a proper codec using audio editing tools.

ALGORITHM:

Step 1: Install tools like audacity, ardour for audio editing, sound mixing and special effects like fade in or fade out etc

Step 2: Perform audio compression by choosing proper codec like mp3, wav, acc etc

Step 3: select and extract the audio file and perform audio editing features to improve the quality of the given codec

Step 4: Extract the completed audio and after modification

Step 5: Now the compressed clean audio is ready to import

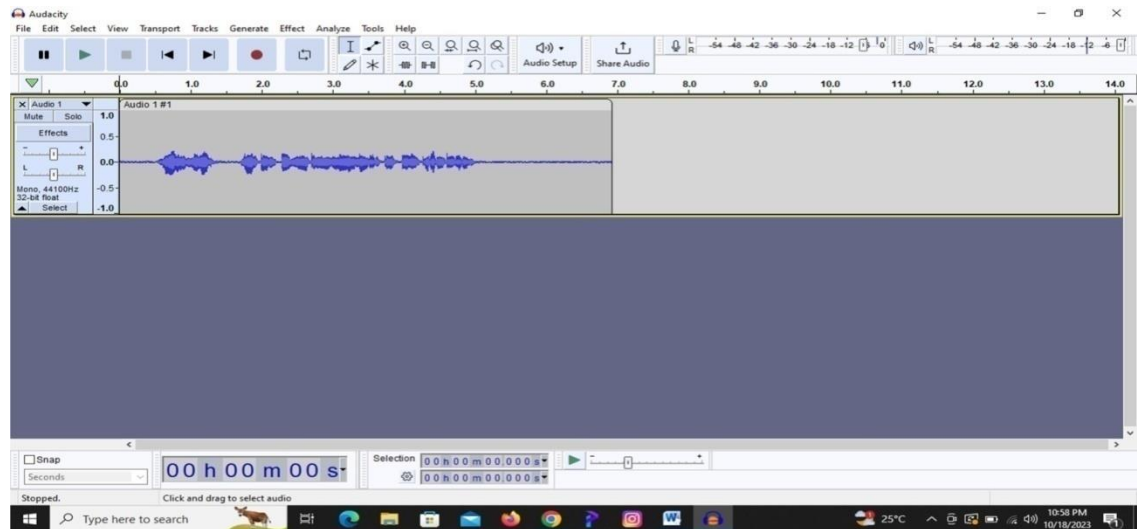
To perform a audio editing first we should do the noise reduction in the background

Then we should compress the imported audio to a feasible one

After doing so we should amplify the audio in case if audio's sound quality seems to be lessened.

PROCEDURES:

Lets first import and record the audio used to perform the compression editing modification

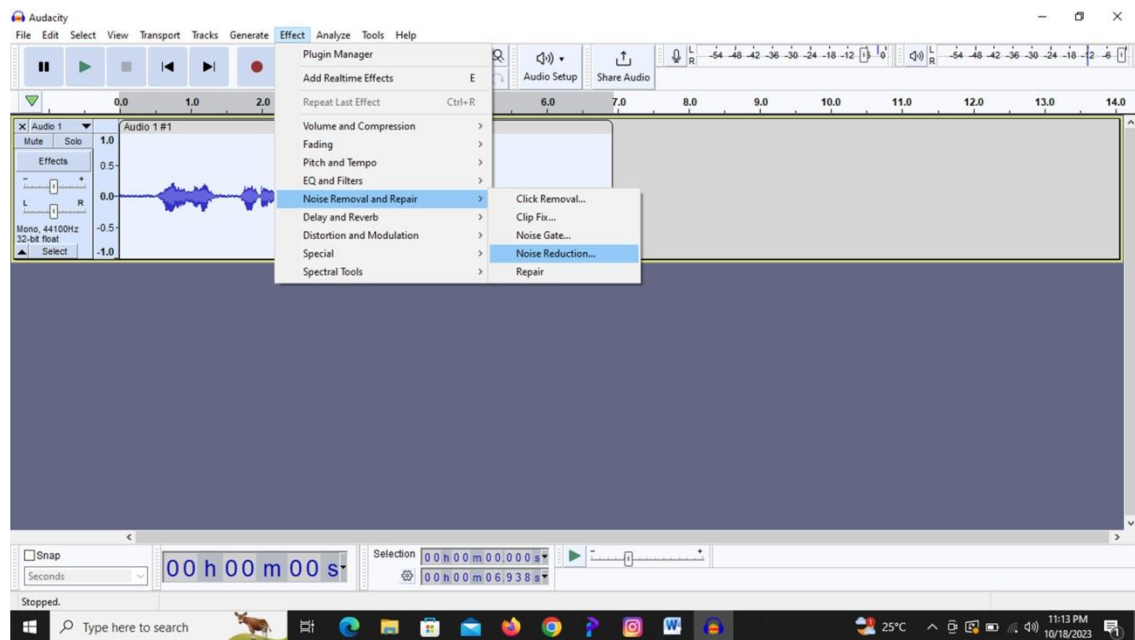


Then after importing the audio select the audio and reduce the noise from the background

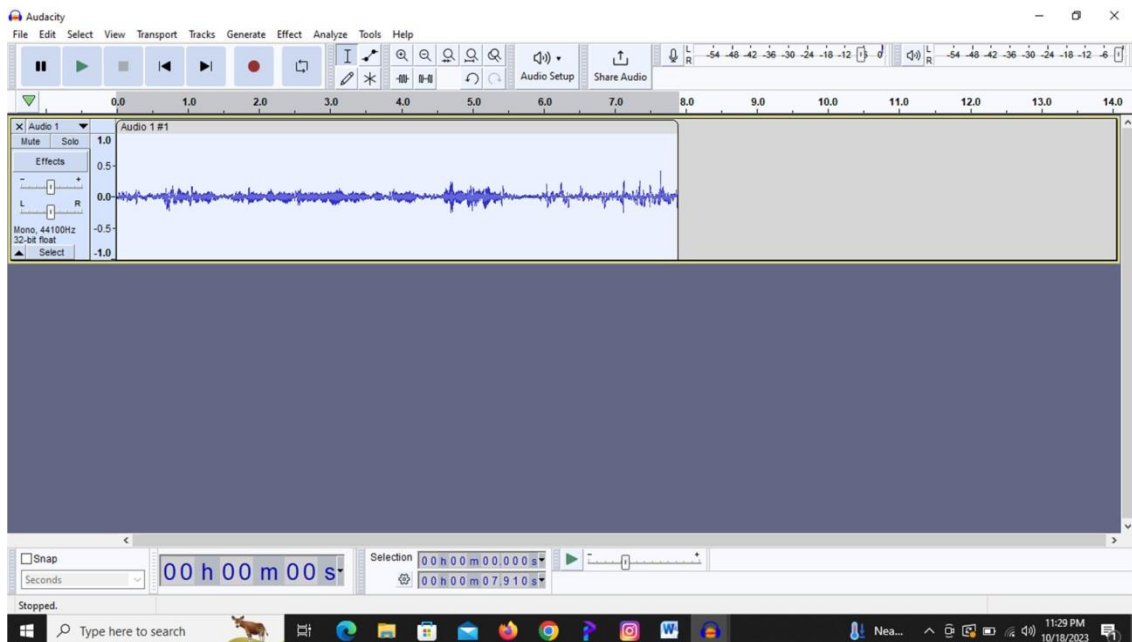
NOISE REDUCTION:

Noise reduction is a crucial audio editing technique used to minimize or remove unwanted background noise from recordings. This can greatly improve the overall quality and clarity of the audio.

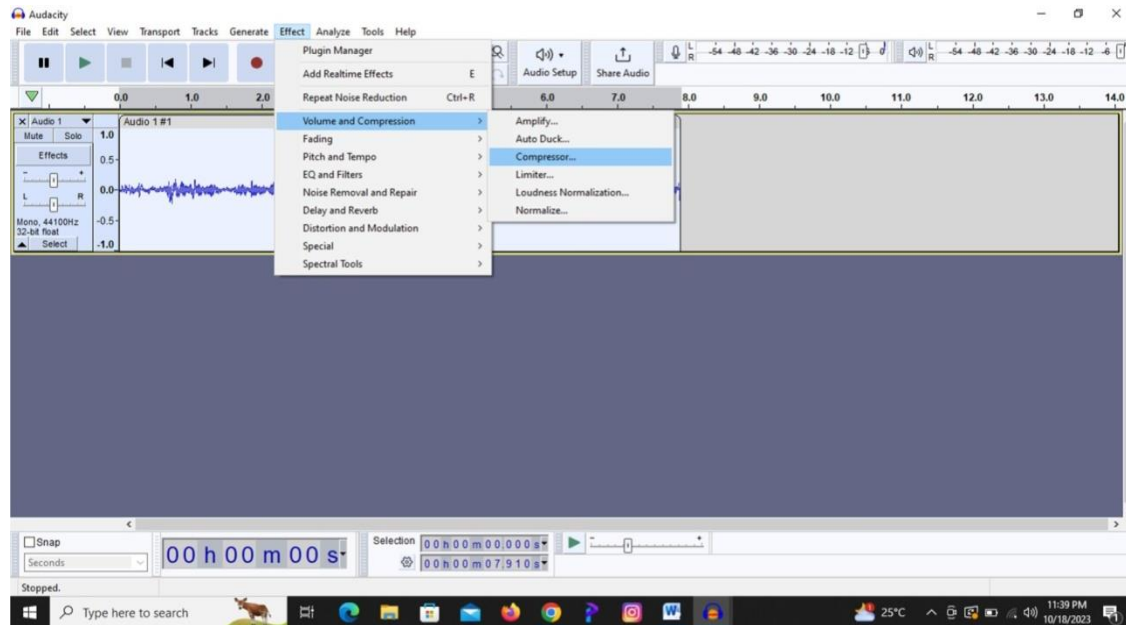
Remember, while noise reduction can significantly improve audio quality, it's important to strike a balance. Over-applying noise reduction can lead to unnatural sounding audio or even the loss of important details. Always listen critically and make adjustments as needed. It's also a good practice to keep a backup of the original audio in case you need to revisit the editing process.



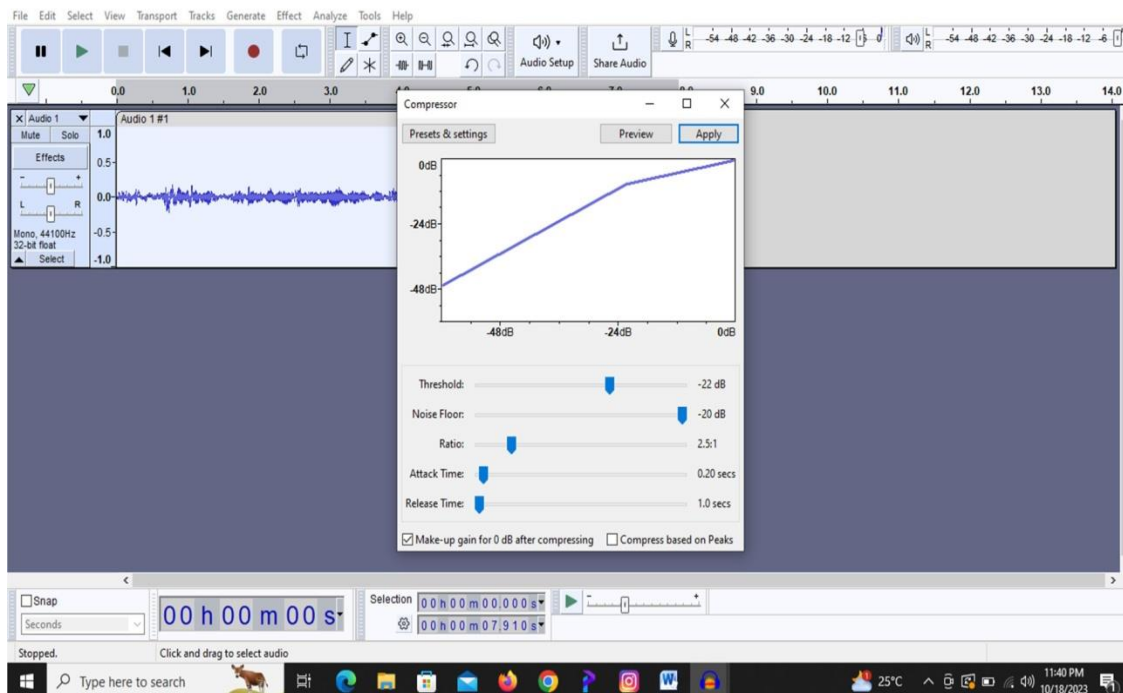
As you can see from the above screen select to effects and select the imported audio then apply noise reduction it will reduce and minimize the unwanted background noise as we can see the noise have been reduced and became to a even audio



After the noise reduction process we can then apply compression to the selected audio

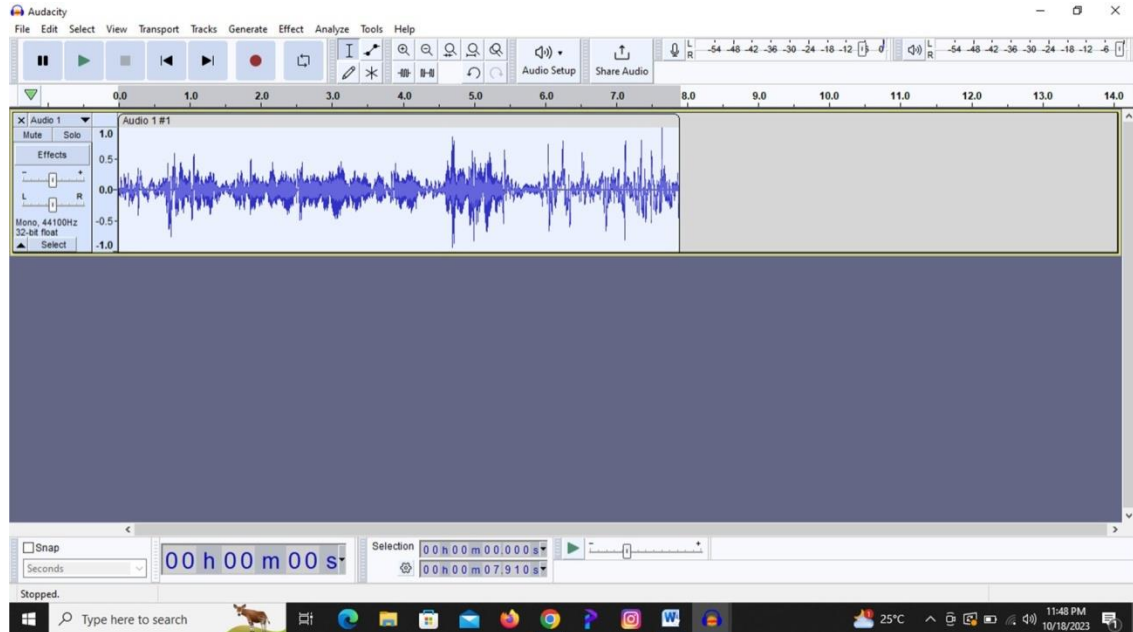


Select and adjust the ratio of components present in the dialog box below as per your require here all are at default percentages like

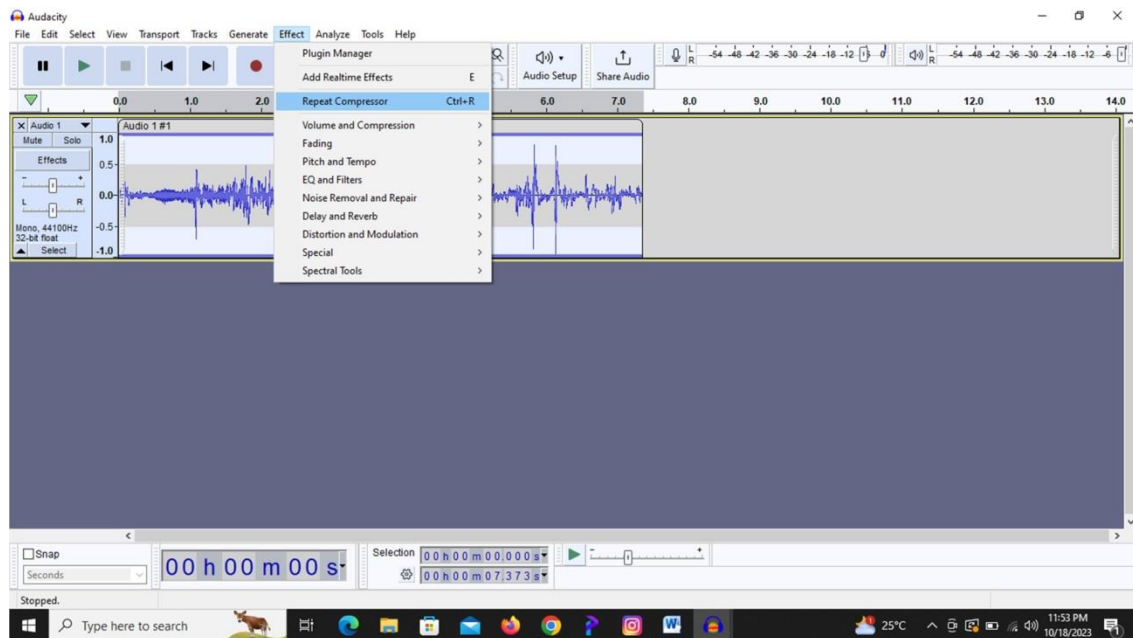


COMPRESSION:

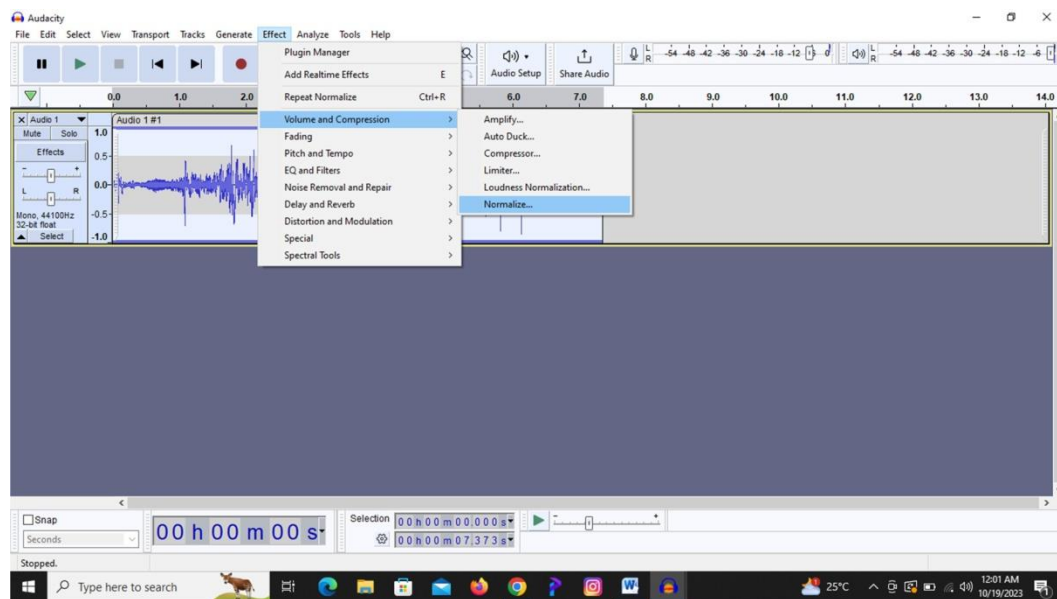
Audio compression is a process used to reduce the dynamic range between the loudest and softest parts of an audio signal.



The compression of the audio have been successful in half way apply we can also repeat compression if we want.

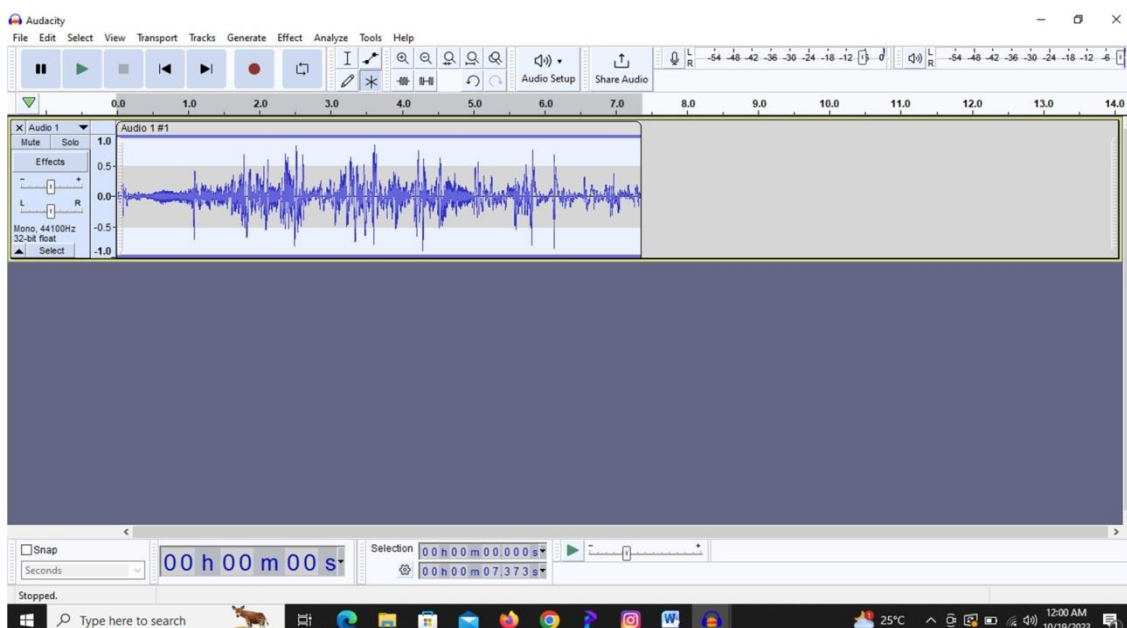


Then we can apply normalization and amplify as a final step to complete the audio editing and enhancement



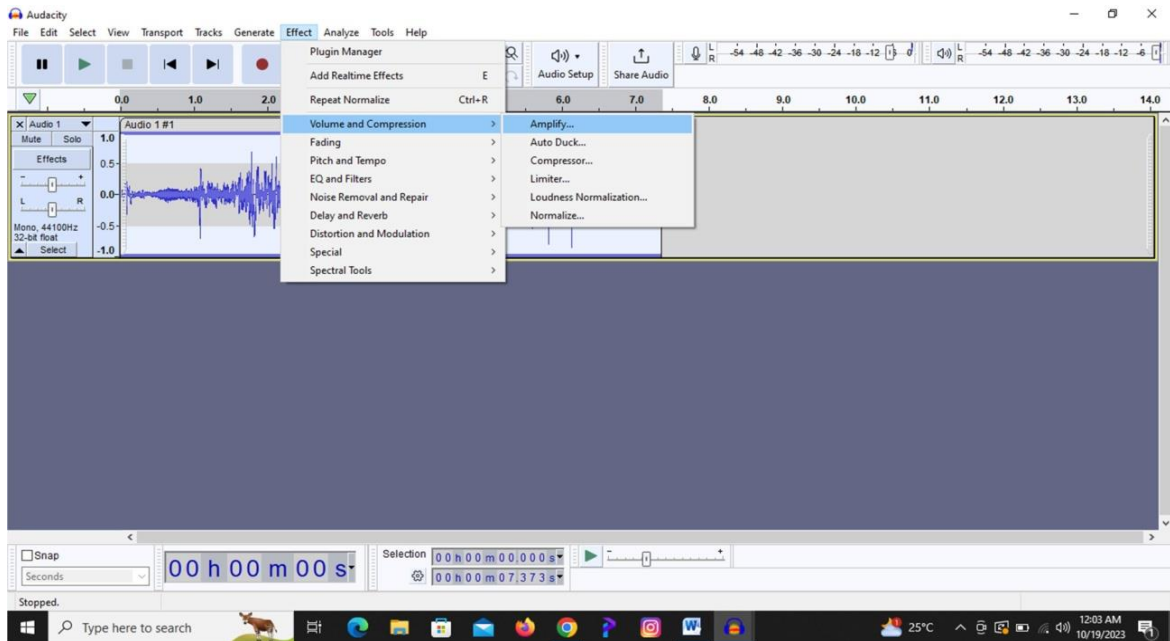
NORMALIZATION:

Normalizing is an audio processing technique used to adjust the overall volume level of an audio track or selection to a specified target level. The goal is to maximize the use of the available dynamic range without causing distortion

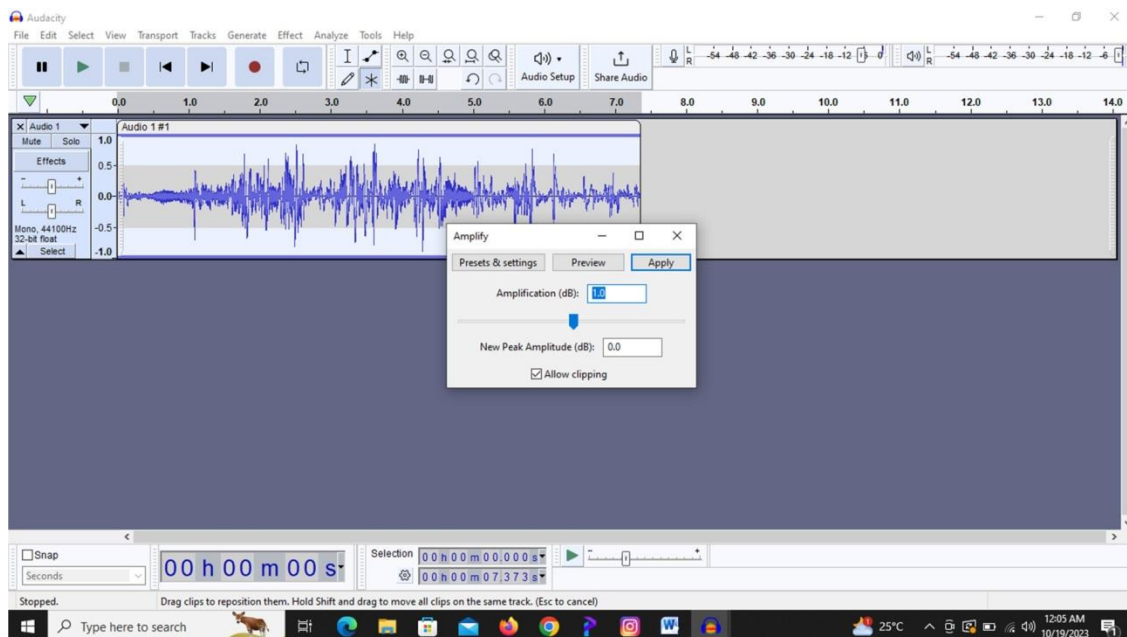


.AMPLIFY:

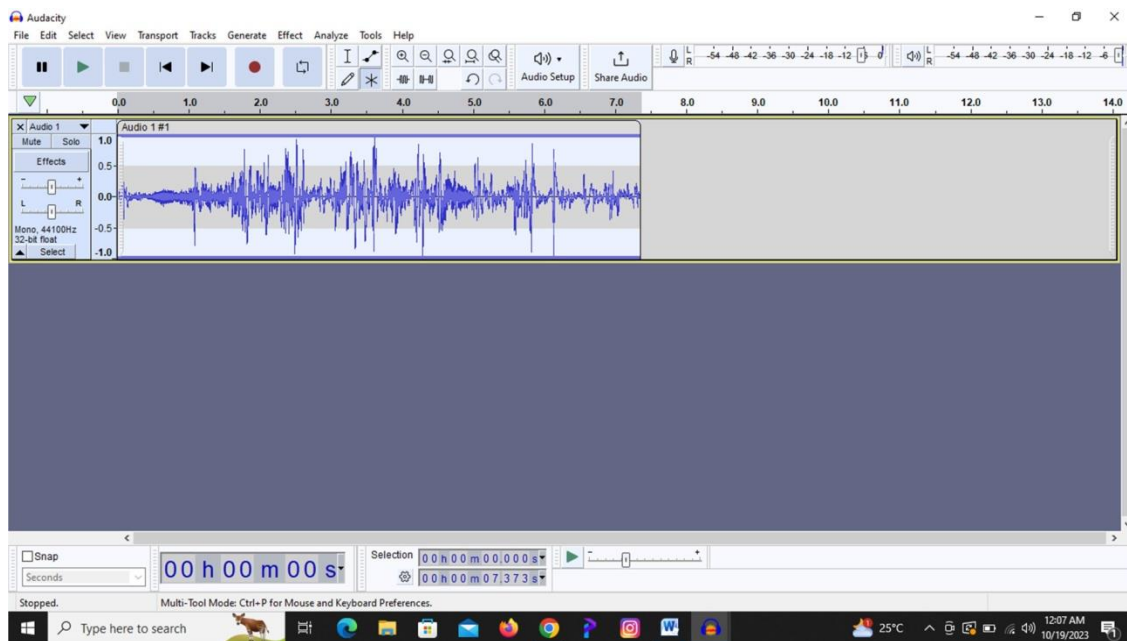
Amplification in audio editing refers to increasing the volume level of a selected portion or the entire audio track. This can be useful to make quiet parts more audible or to adjust the overall volume to a desired level. In Audacity, here are the steps to amplify audio.



You can select any amplify rates dbs u want to synchronize



At final our audio is ready to export with audio edited features using proper codec



RESULT:

Thus the Conversion of Audio Compression Using Proper Codec was Implemented And Executed Successfully.

EX.NO : 3	WORKING WITH VIDEO EDITING TOOLS
DATE :	

AIM:

To Edit and mix video content, remove noise, create special effects and captions using Openshot.

ALGORITHM:

Step 1 :Import Photos & Music.

Step 2 : Add Photos to Timeline.

Step 3 : Add Music to Timeline.

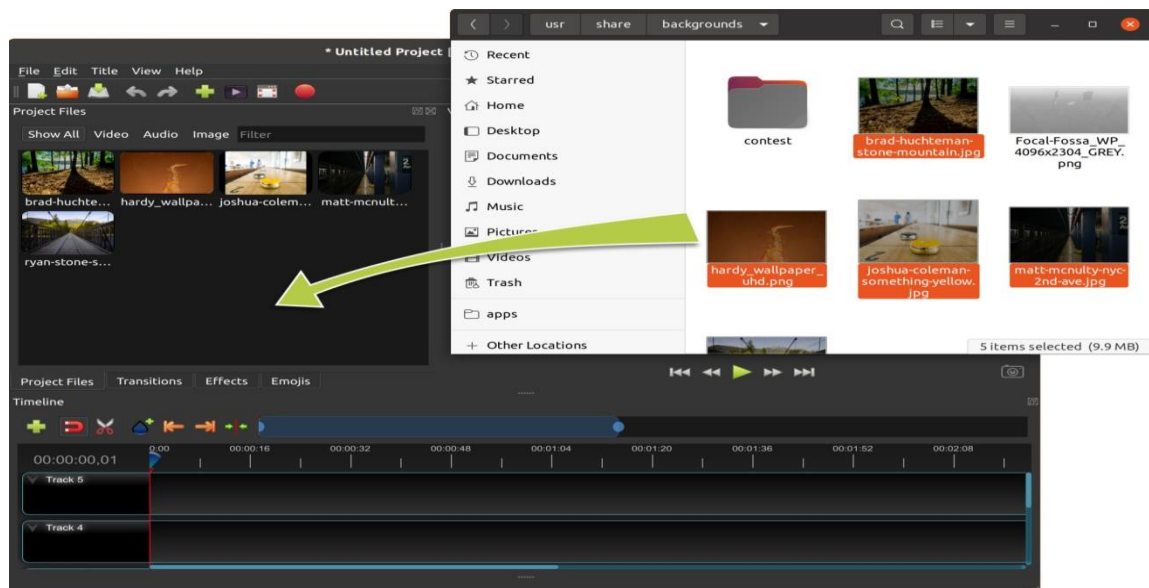
Step 4 : Preview your Project.

Step 5 : Export your Video.

PROCEDURE:

Step 1 :Import Photos & Music

Before we can begin making a video, we need to import media files into OpenShot. Most video, image and music file formats will work. Drag and drop a few videos or images and a music file from your Desktop to OpenShot. Be sure to drop the files where the arrow in the illustration is pointing to.

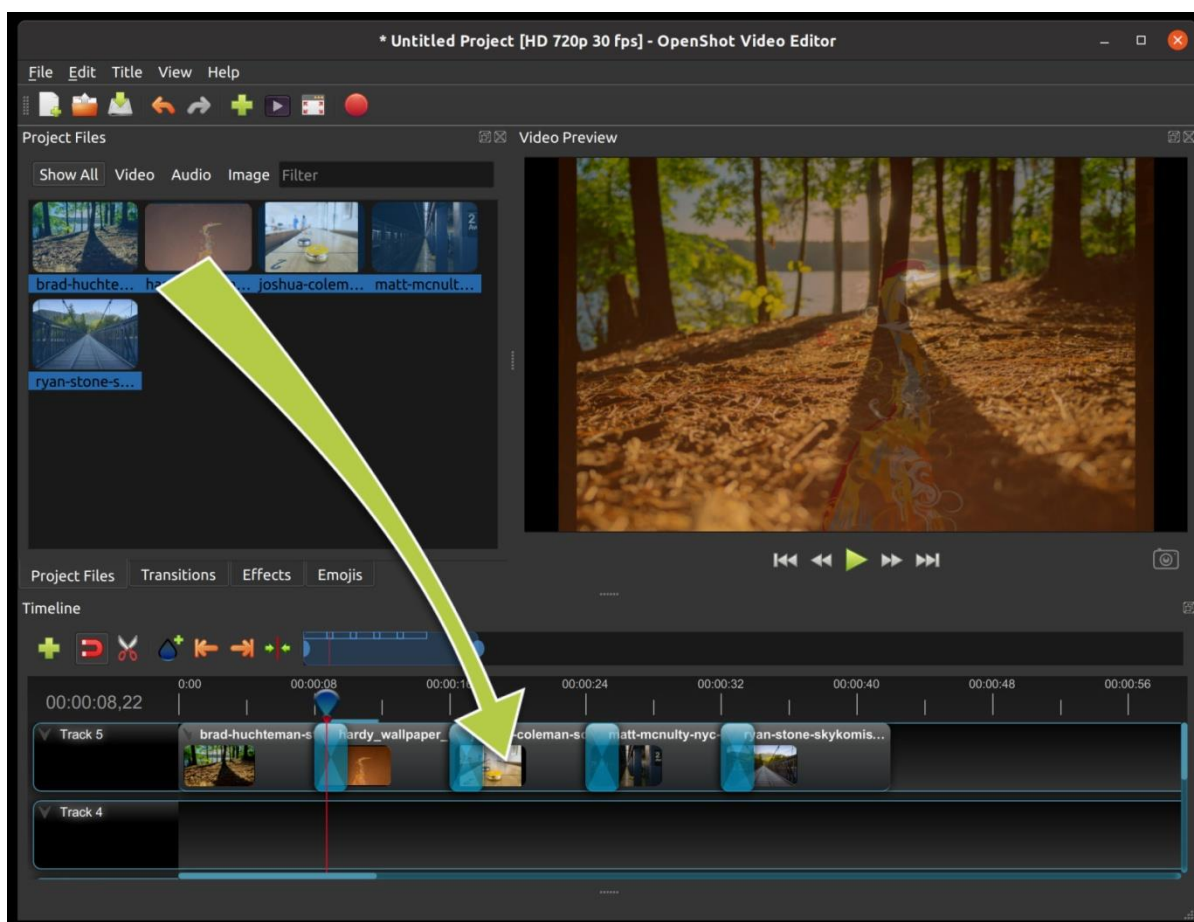


Alternative methods to add files to your projects are described in the section [Import Files](#). The “Show All”, “Video”, “Audio”, “Image” filters above the added files allows you to only see the file types you are interested in.

Step 2 :Add Photos to Timeline

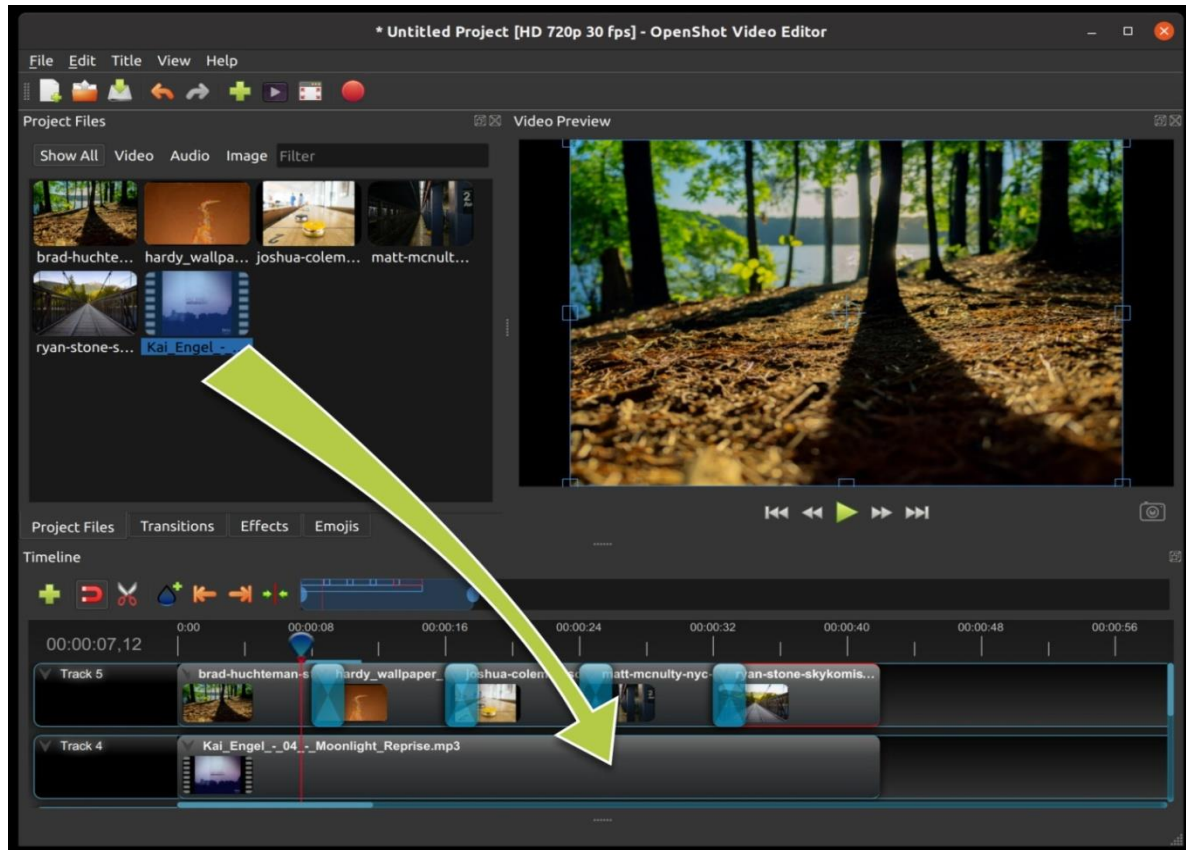
Next, drag each video or photo onto a track in the timeline (as seen in the illustration). The timeline represents your final video, so arrange your photos (i.e. clips) in whatever sequence you want them to appear in your video. If you overlap two clips, OpenShot will automatically create a smooth fade between them (only affects the image, and not audio), displayed by blue rounded rectangles between the clips. Remember, you can rearrange the clips as many times as needed by simply dragging and dropping them.

You can also shorten or lengthen each clip, by clicking the left or right edge and dragging your mouse. For example, if you want a photo to last longer than 10 seconds (the default duration), simply grab the right edge of the photo (on the timeline), and drag it to the right (to increase the clip’s duration on the timeline).



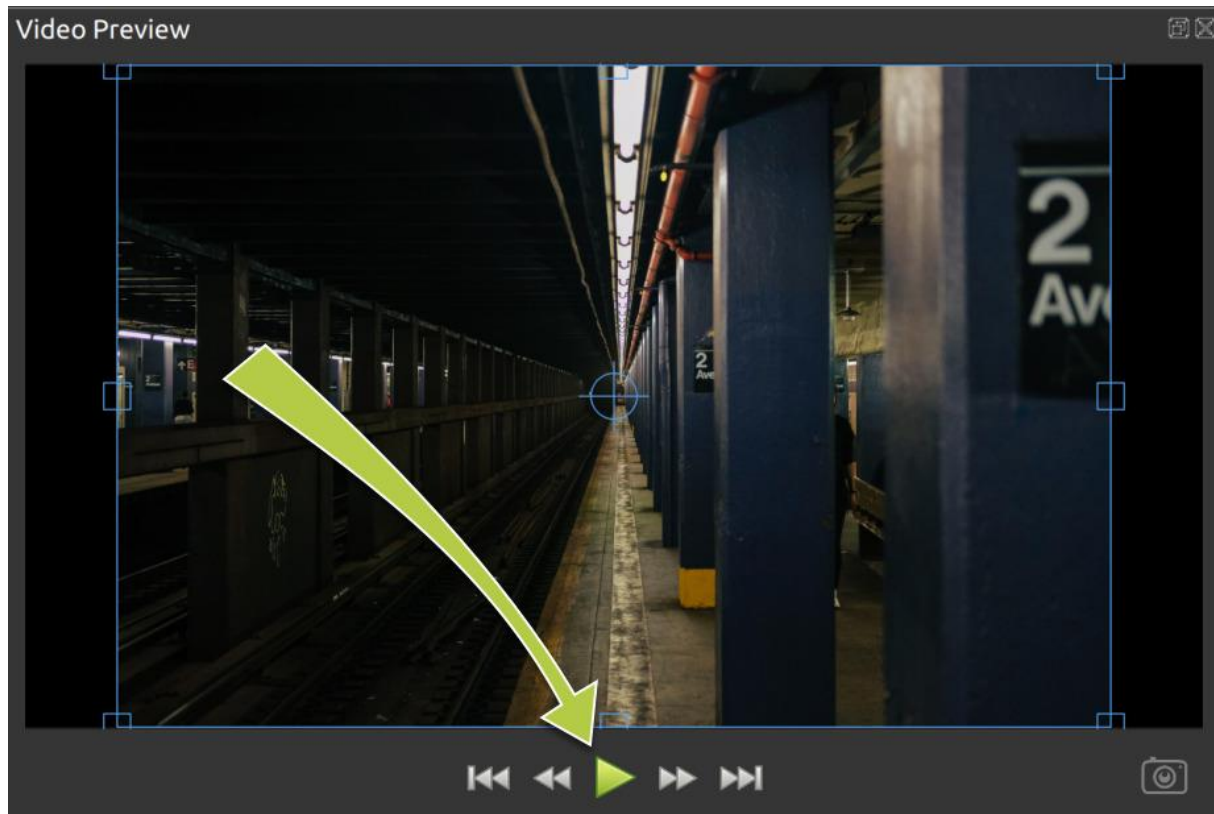
Step 3 : Add Music to Timeline

To make our creation more interesting, we need to add some music. Click on the music file that you imported in step 1, and drag it onto the timeline. If the song is too long, grab the right edge of your music clip, and resize it smaller (that will make it end earlier). You could also insert the same file multiple times, if your music is too short.



Step 4 : Preview your Project

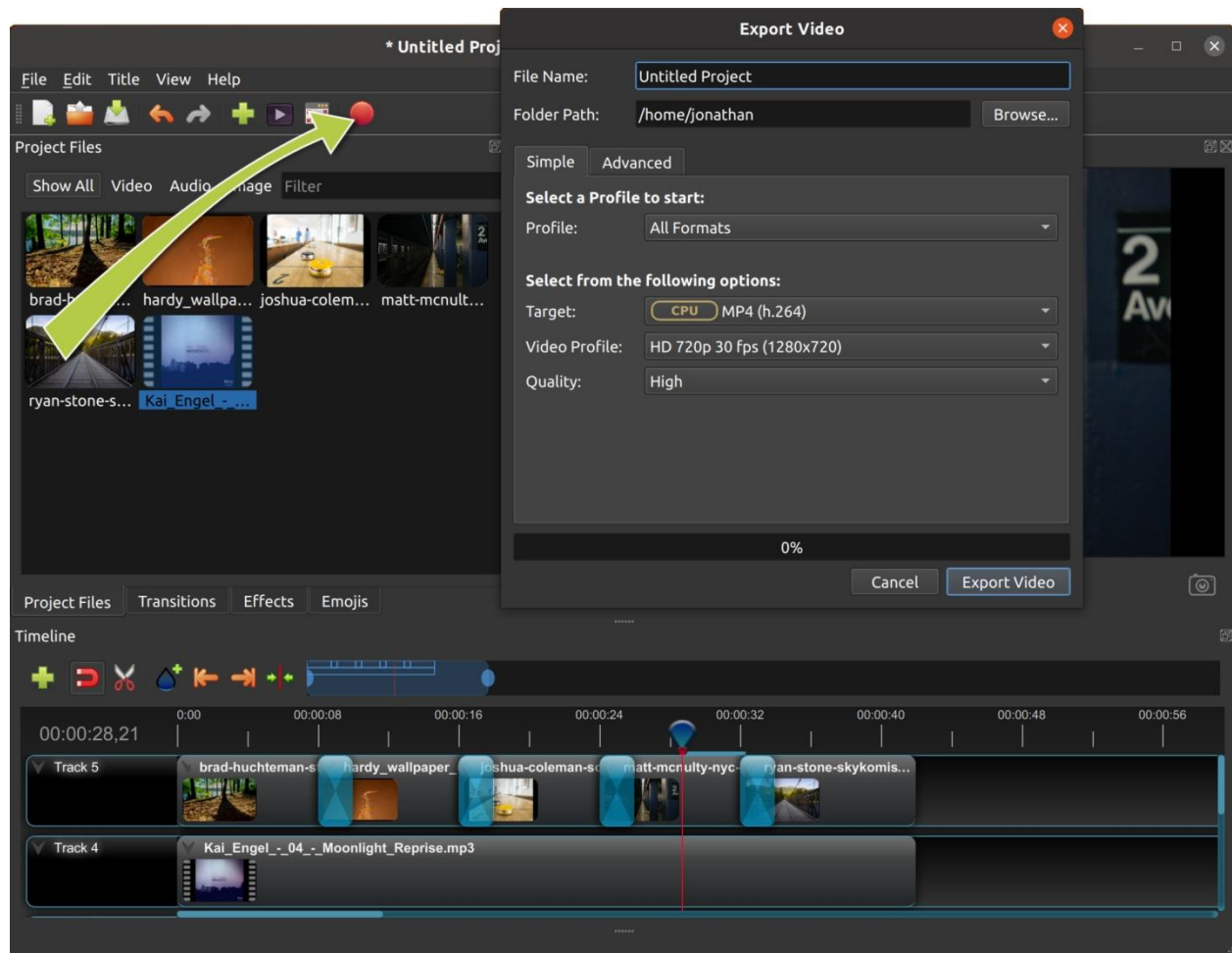
To preview what our video looks & sounds like, click the Play button under the preview window. You can also pause, rewind, and fast-forward your video project by clicking the corresponding buttons.



Step 5 :Export your Video

Once you have edited your photo slide-show video, the last step is to export the project. Exporting converts your OpenShot project into a single video output file. By using the default settings, the video works on most media players (such as VLC) or websites (such as YouTube, Vimeo, ...).

Click on the Export Video icon at the top of the screen (or use the [File→Export Video](#) menu). The default values will work fine, so just click the [Export Video](#) button to render your new video.



RESULT:

Thus the Edit and mix video content, remove noise, create special effects and captions using Openshot have been executed and verified Successfully.

EX.NO : 4	WORKING WITH WEB/MOBILE AUTHORING TOOLS
DATE :	

a) Design simple homepage with banners, logos, tables, quick links etc.

AIM :

To design simple homepage with banners, logos, tables, quick links etc.

ALGORITHM:

Step 1: Start the program

Step2: Install visual studio code and create new folder.

Step3: Create a new file and save it with html and css extensions.

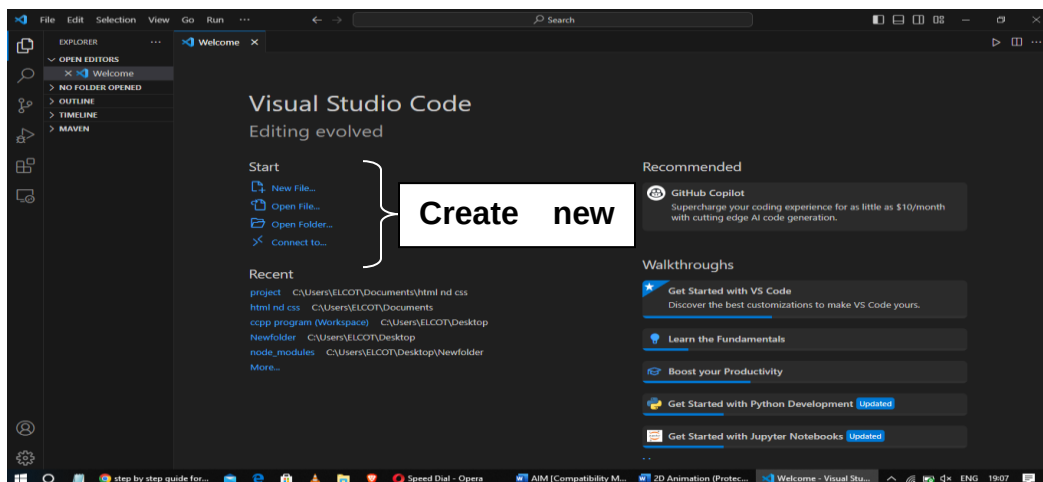
Step4: Create a simple homepage.

Step5: Add banner, logos, tables quick links etc.,

Step6: Stop the program.

PROCEDURE:

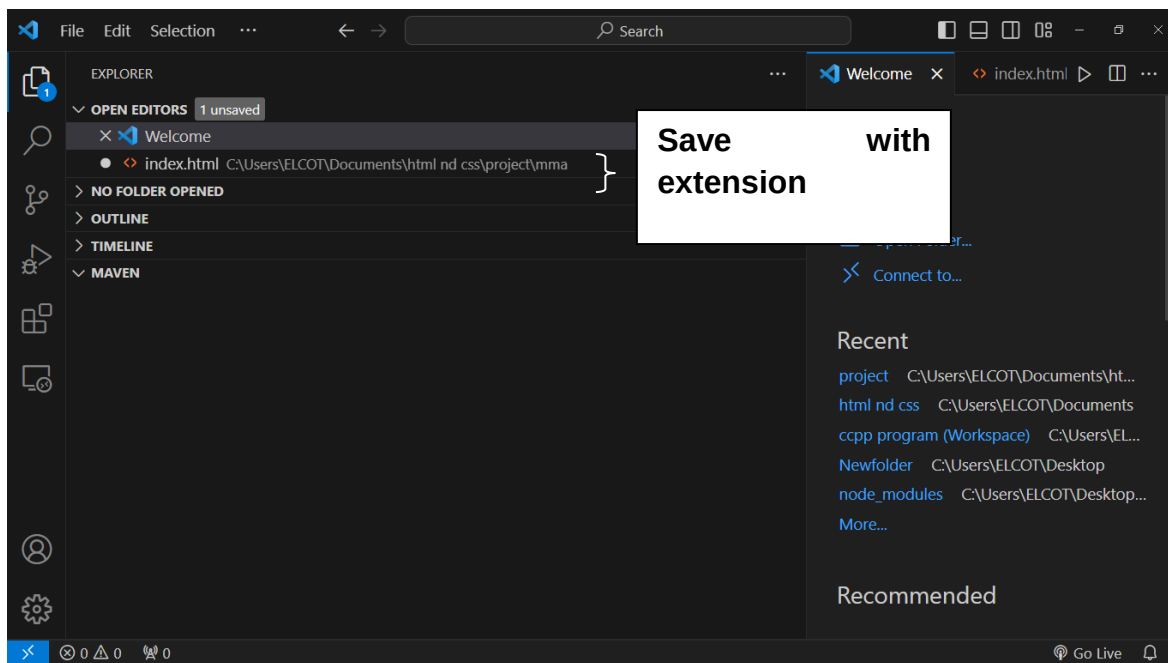
Step1: Install visual studio code.



Step2:

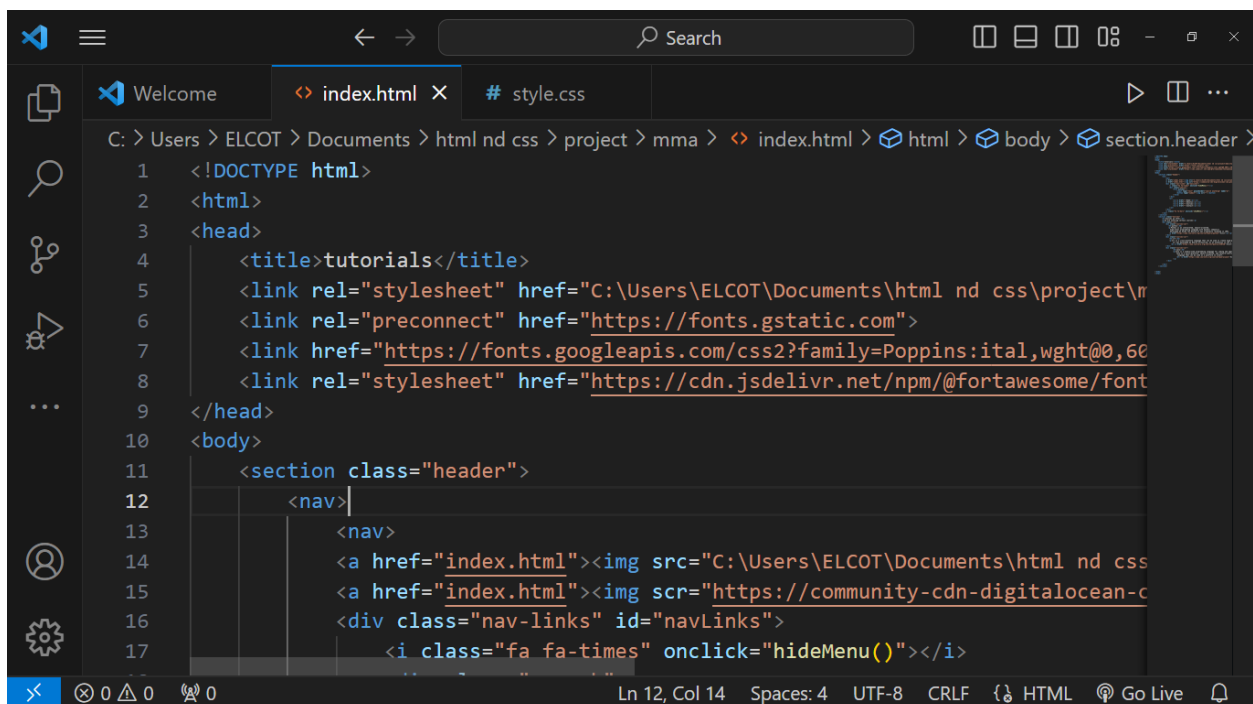
- Choose File > Save As and choose HTML from the drop-down list.
- Give the filename an extension of .html, specify the file location, and click Save.

c) Open the HTML file in a Web browser to examine the converted file. If it meets with your approval, you are done.

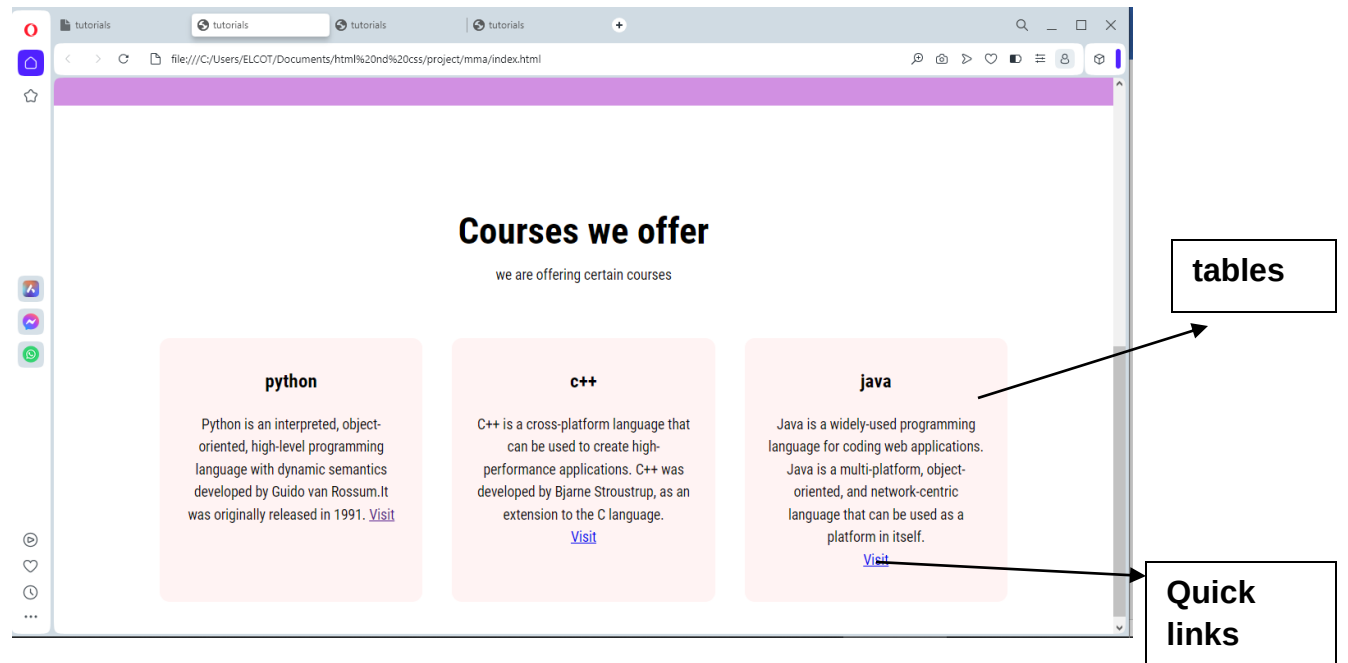


Step3: Create simple home page using the html and css code.

Step4: Add banner, logos, tables quick links to your home page.



OUTPUT:



RESULT:

Thus the design of simple homepage with banners, logos, tables, quick links etc created and verified Successfully.

b) Provide a search interface and simple navigation from the homepage to the inside page of the website.

AIM:

To provide a search interface and simple navigation from the homepage to the inside page of the website.

ALGORITHM:

Step1: Start the program.

Step2: Insert the search interface to the webpage.

Step3: Add the navigation bar from the home page to the inside page of the website.

Step4: Stop the program.

PROCEDURE:

In the previous section, we created the structure of the basic site where we are going to use the Navigation bar with the search bar with the icon. We will design the structure and attach the icons for each navbar

Step1:

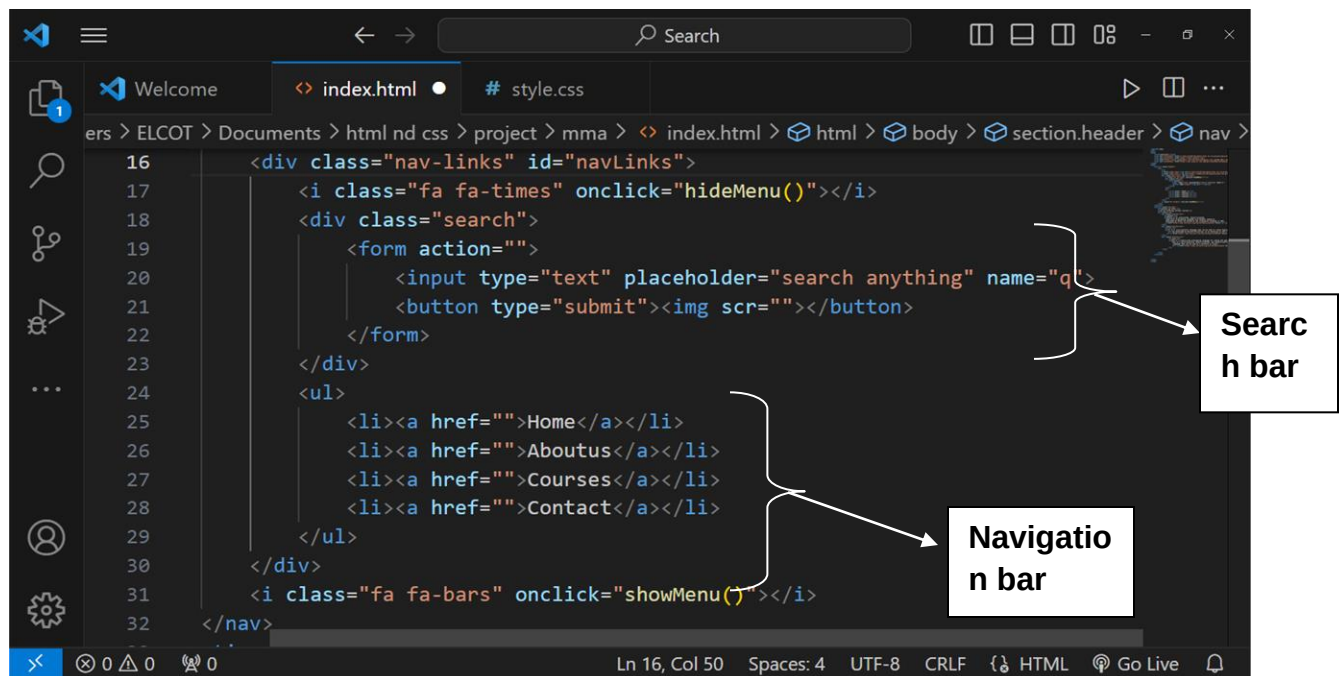
The HTML code is used to create a structure of navigation bar containing search bar. We will use some CSS property to make it attractive

Step2:

CSS code is used to make the attractive website. This CSS property is used to make the style on navigation bar containing search bar.

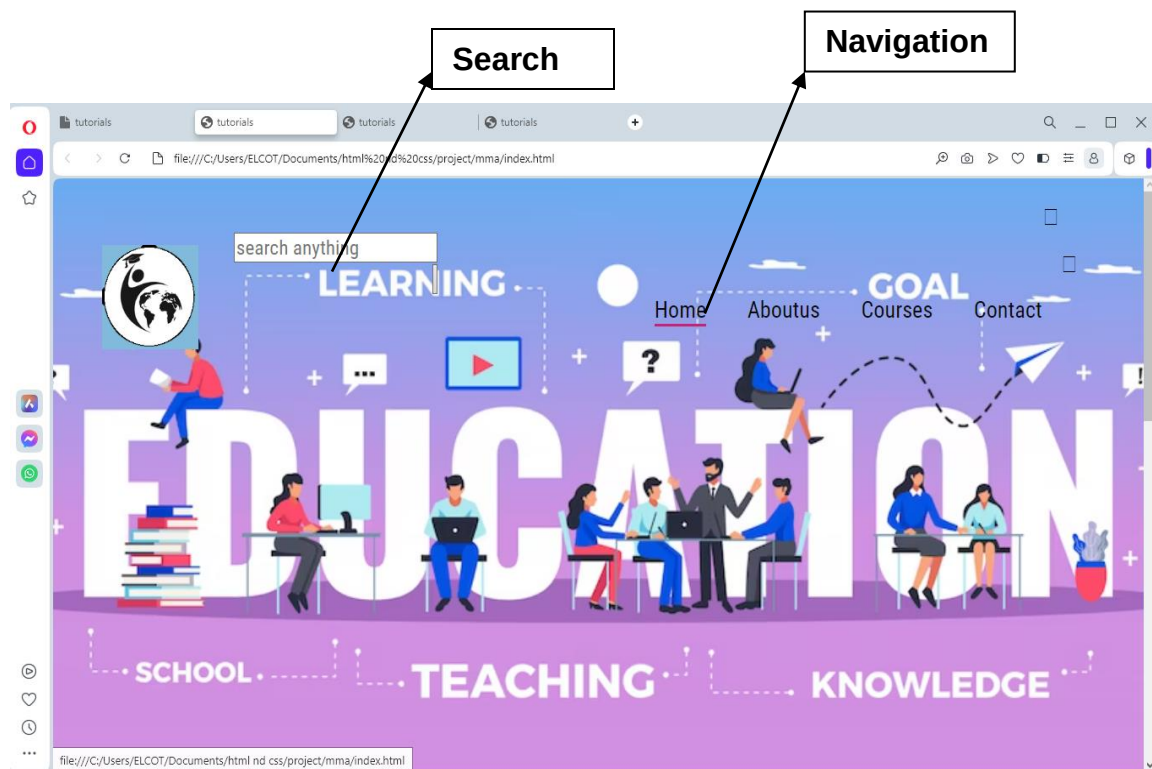
Step3:

A navigation bar is basically a list of links, so using the and elements makes perfect sense



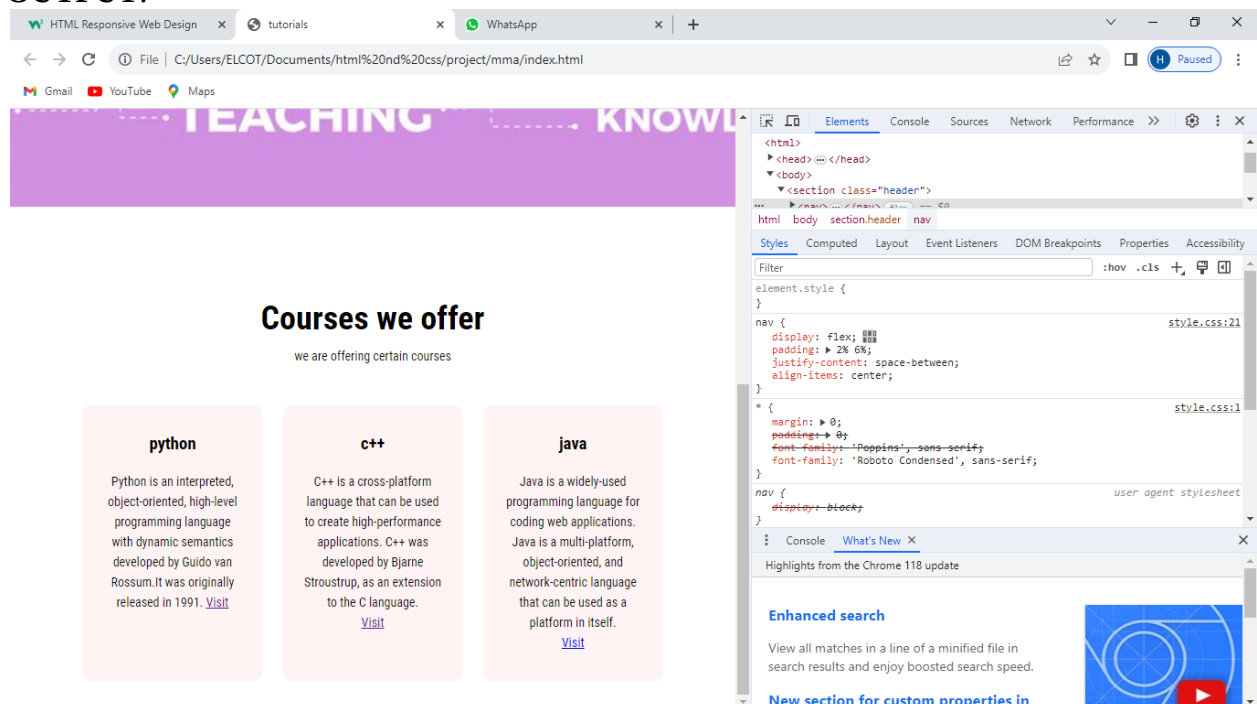
The screenshot shows the VS Code editor with a file named `index.html` open. The code defines a navigation bar and a search bar. The navigation bar is a `<nav>` element containing a `<div class="nav-links" id="navLinks">` which has a `<i class="fa fa-times" onclick="hideMenu()"></i>` and a `<ul>` with links for Home, Aboutus, Courses, and Contact. The search bar is a `<div class="search">` containing a `<form>` with a text input (placeholder "search anything", name "q") and a submit button. Annotations with arrows point from the text boxes "Search h bar" and "Navigation n bar" to their respective code blocks.

```
16 <div class="nav-links" id="navLinks">
17   <i class="fa fa-times" onclick="hideMenu()"></i>
18   <div class="search">
19     <form action="">
20       <input type="text" placeholder="search anything" name="q">
21       <button type="submit"><img scr=""></button>
22     </form>
23   </div>
24   <ul>
25     <li><a href="">Home</a></li>
26     <li><a href="">Aboutus</a></li>
27     <li><a href="">Courses</a></li>
28     <li><a href="">Contact</a></li>
29   </ul>
30 </div>
31 <i class="fa fa-bars" onclick="showMenu()"></i>
32 </nav>
```



RESULT:

Thus the search interface and simple navigation from the homepage to the inside page of the website is provided and verified Successfully.



RESULT:

Thus the design of Responsive web pages for use on both web and mobile interfaces was created and verified Successfully.

EX.NO : 5 a	WORKING WITH ANIMATION TOOLS
DATE :	

AIM:

To Perform a simple 2D animation using krita software and wick editor.

ALGORITHM:

Step 1: Creating a separated 2D model images by using the krita software.

Step 2: Using the sprites tools to adding green screen effect to separate 2D models .

Step 3: Import the sprites images in the Wick Editor software.

Step 4: arrange images in the first frame.

Step 5: Add movements in all frames by onion skinning tool.

Step 6: Export the video by export feature.

PROCEDURE:

Creating an 2D images by using the krita

Step 1: Create a New Canvas

Click on "File" > "New" or use the keyboard shortcut (Ctrl + N / Command + N).

In the dialog box that appears, set the canvas size. For sprites, this can be relatively small, depending on the intended use. Common sprite sizes are 32x32, 64x64, or 128x128 pixels.

Set the resolution and color mode as per your project requirements.

Click "Create" to generate the canvas.

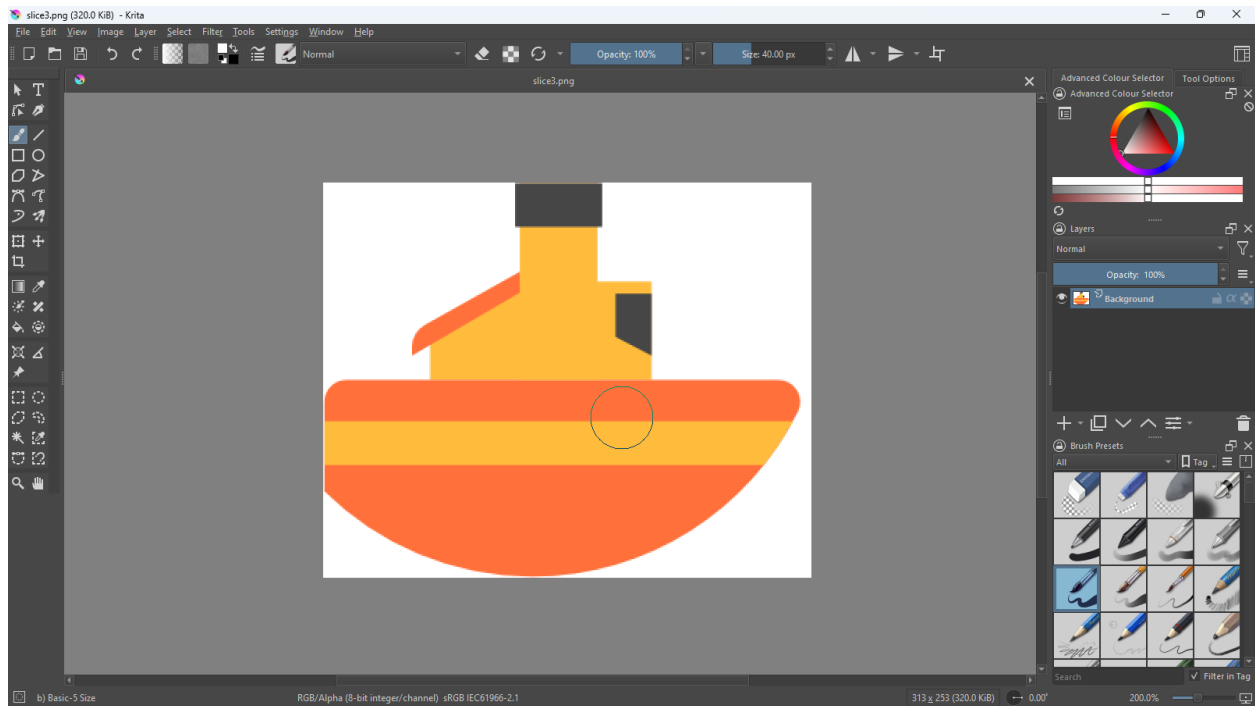
Step 2: Designing an images:

Use the various drawing tools in Krita to create your sprite. The most common tools include the Brush, Pencil, and Line tools.

You can also use layers to separate different elements of your sprite, making it easier to edit and manage.

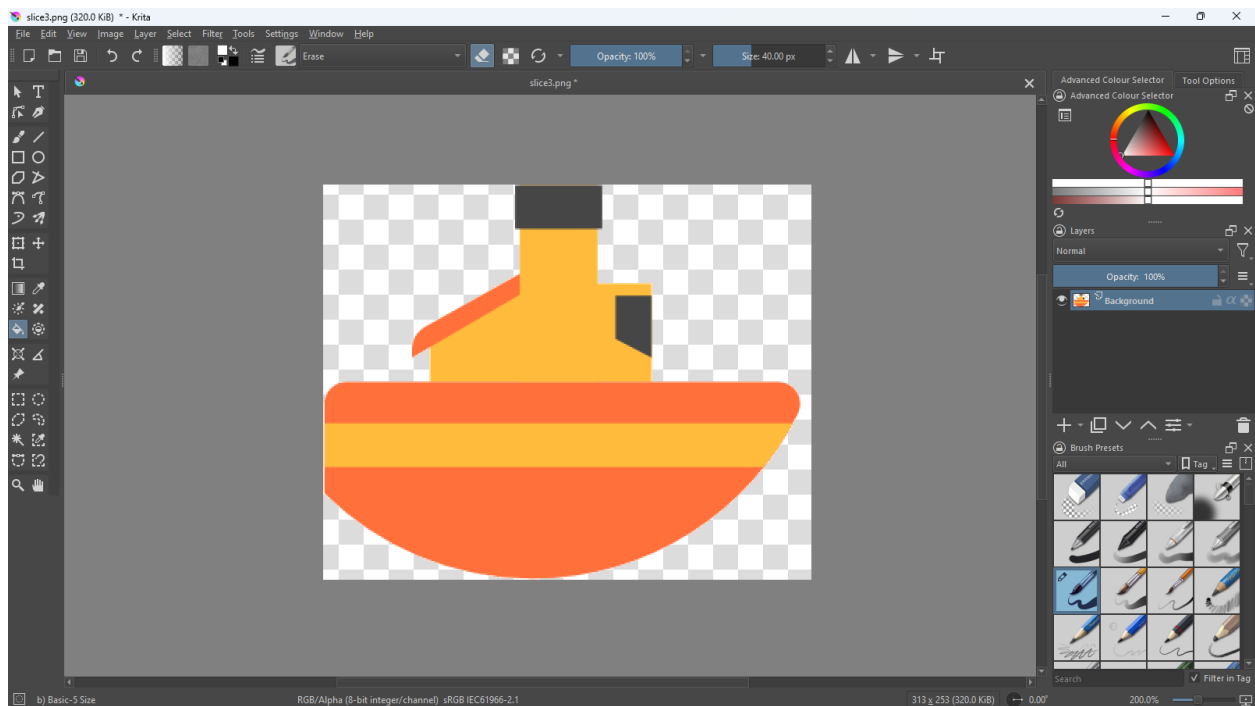
Step 3: Coloring an images:

Use the color palette, which you can access in the "Advanced Color Selector" docker or the "Color" menu, to choose and apply colors to your image.



Step 4: Sprites an images:

Use the sprites tool to add green screen in the images



Step 5: Export Your images:

Once you've completed your sprite, you'll need to export it in a format suitable for your specific project. Common sprite file formats include PNG or GIF.

Go to "File" > "Export" and choose your desired format.

CREATE 2D ANIMATION IN WICK EDITOR

Step 1: Create a New Project

Click on "File" and select "New Project" to create a new project. Give your project a name and choose a location to save it.

Step 2: Create a New Scene

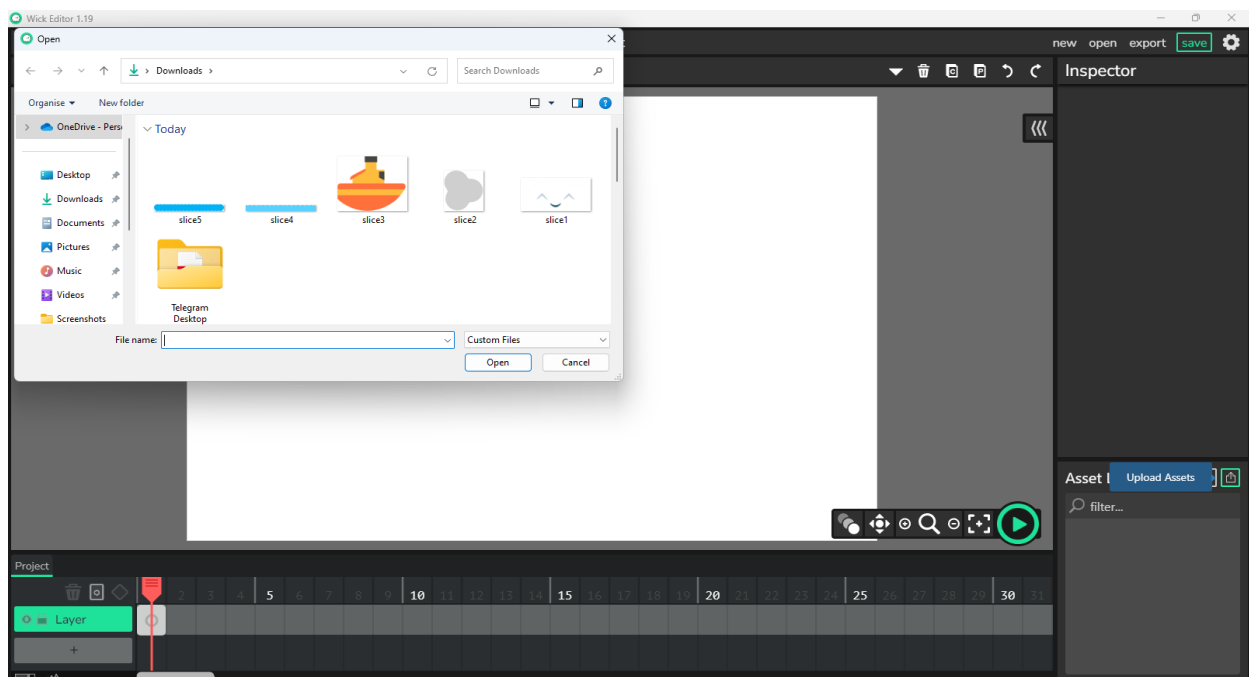
Click on "New Scene" to add a new scene to your project. Scenes are like frames in traditional animation.

Step 3: Add and Edit Objects

To create an animation, you'll need to add objects to your scene. Click the "New Folder" button in the Library panel to create a new folder for organizing your objects.

Inside the folder, click the "New" button to add objects like shapes, images, or text.

Select an object, and you can use the Timeline and Property panel to adjust the object's properties and animation properties.



Step 4: Animate Your Objects

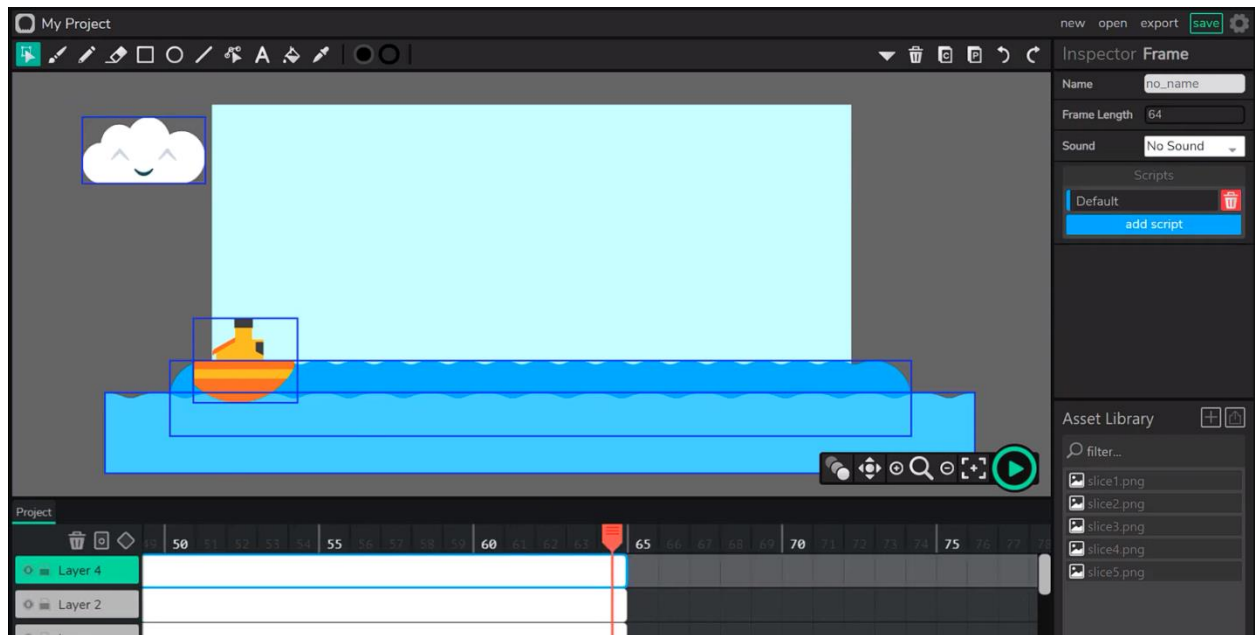
To animate objects, you can create keyframes at different points in the Timeline by clicking on the "Add Keyframe" button.

Move the Timeline playhead to a different frame, make changes to the object's properties (position, size, rotation, etc.), and then add another keyframe. Wick Editor will automatically interpolate the animation between keyframes.

Continue adding keyframes and making changes to create smooth animations.

Step 5: Add Motion Tweens

You can also create motion tweens to simplify the animation process. Select an object, right-click on the frame in the Timeline, and choose "Create Motion Tween." This allows you to move an object from one position to another, and Wick Editor will generate the in-between frames for you.

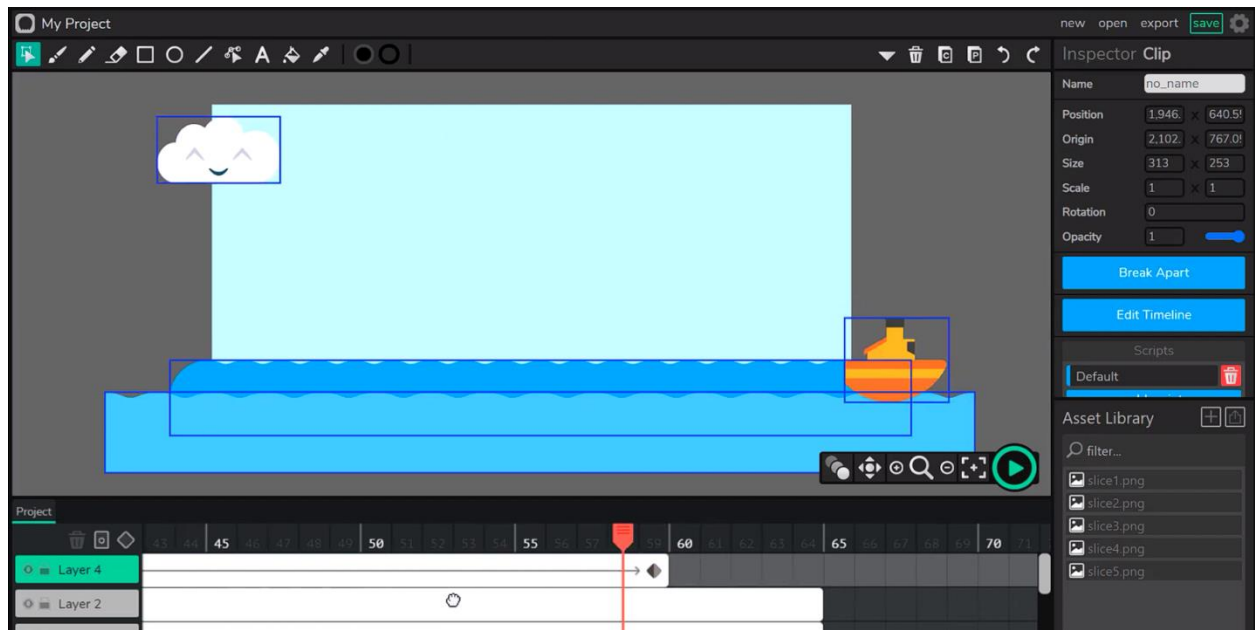


Step 6: Add Sound

To add sound to your animation, click the "Library" panel, and then the "Sounds" folder. You can upload or record sound files to add to your project.

Step 7: Preview Your Animation

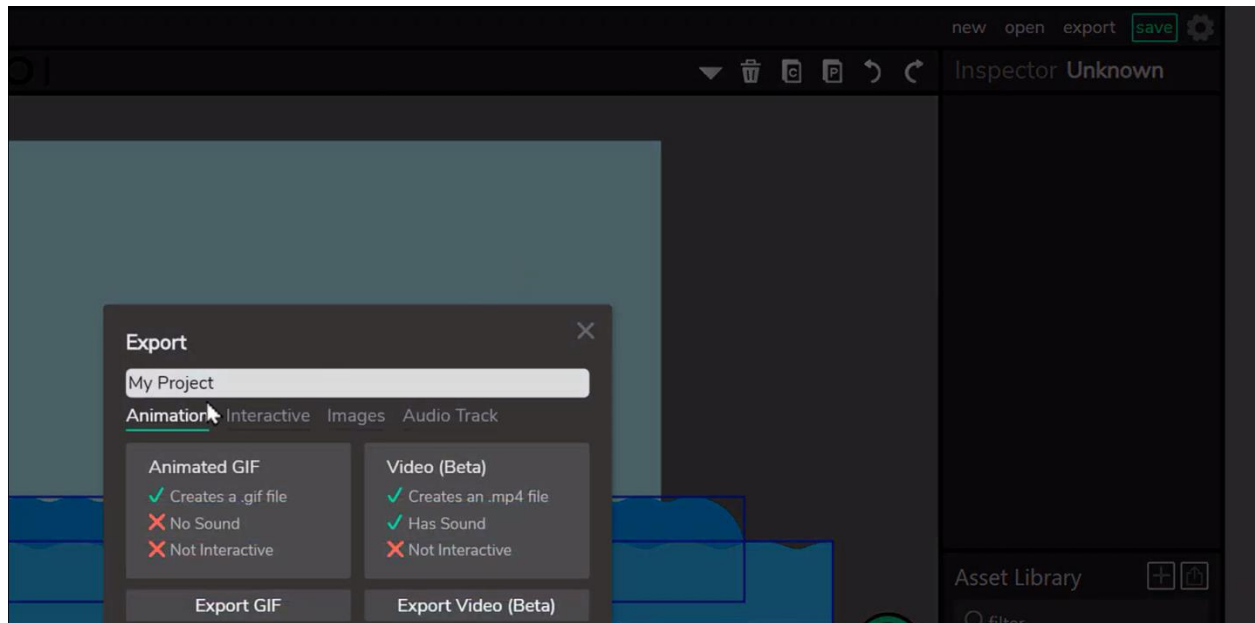
Click the "Play" button in the Timeline to preview your animation. Make adjustments as needed to achieve the desired look and timing.



Step 8: Export Your Animation

Once you're satisfied with your animation, click on "File" and select "Export." Choose your preferred export format (e.g., GIF, video), and configure export settings.

Click "Export" to save your animation to your computer.



RESULT:

Thus the 2D animation using krita software and wick editor has been created and verified successfully

.

EX.NO : 5 b	WORKING WITH ANIMATION TOOLS
DATE :	

AIM:

To Perform simple 3D animation.

ALGORITHM:

Step 1: Creating model.

Step 2: Coloring the model.

Step 3: Adding light and camera.

Step 4: Adding movement to the model.

Step 5: Capture the movement of a model by using camera.

Step 6: Rendering the output.

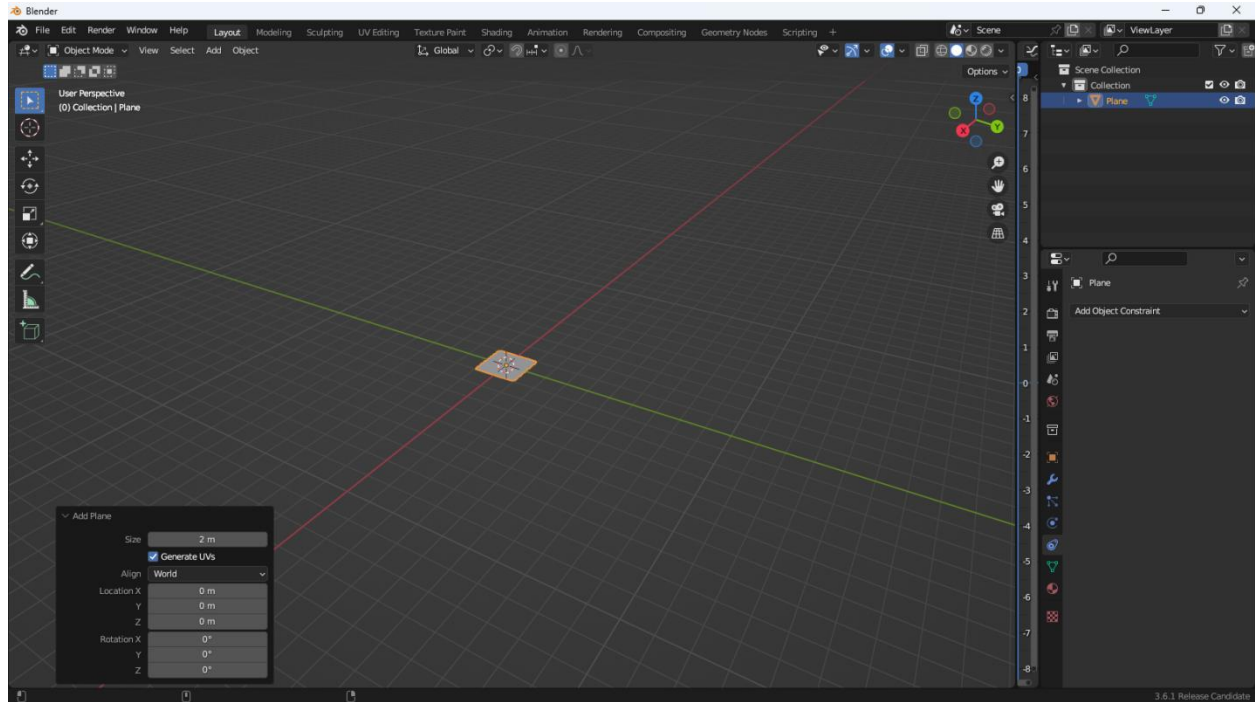
PROCEDURE:

Step 1: Create Your Scene

Launch Blender.

Start by deleting the default cube, lamp, and camera objects (right-click on each and press "Delete" or "X").

Create the 3D scene by adding objects (Shift + A) like models, characters, and props. You can also import 3D models if needed.



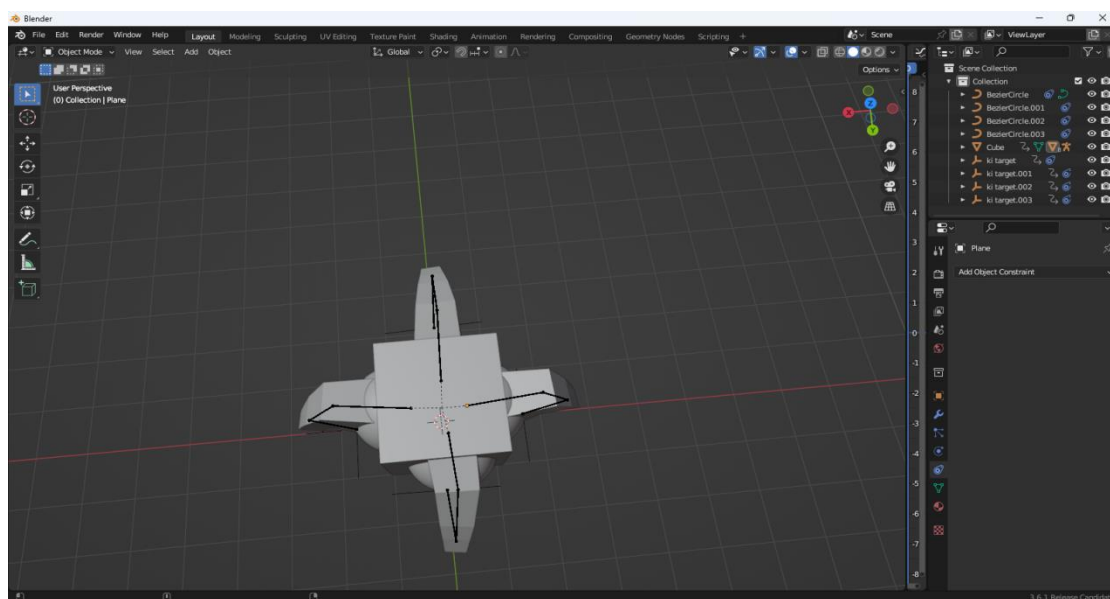
Step 2: Set Up Your Animation

Create a model.

Select the object you want to animate by right-clicking it.

Go to the "Layout" workspace or "Animation" workspace to work on your animation.

In the timeline at the bottom, click the "Record" button (a red circle) to enable auto-keyframing. This will automatically create keyframes when you make changes to the object.

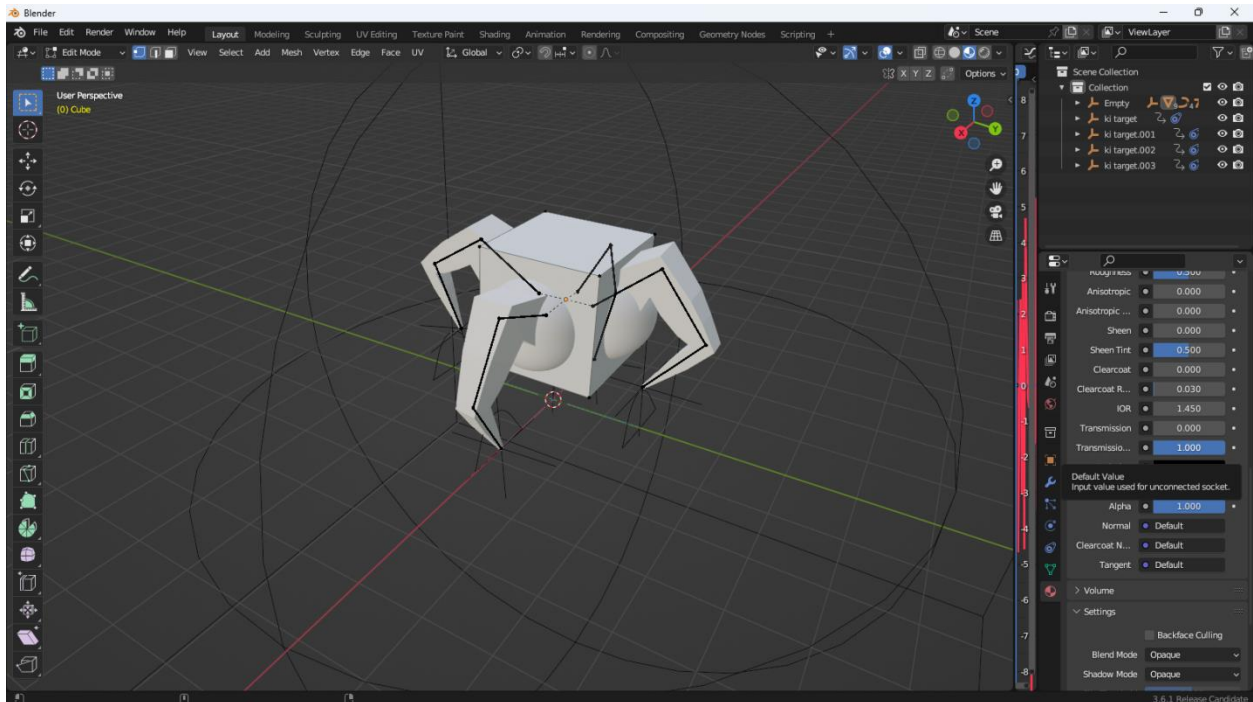


Step 3: Keyframing

Move to the frame where you want to start your animation (e.g., frame 1).

Adjust the object's position, rotation, or scale as desired.

Press "I" on your keyboard, and choose the type of keyframe (Location, Rotation, Scale, etc.) to set a keyframe for the selected property.



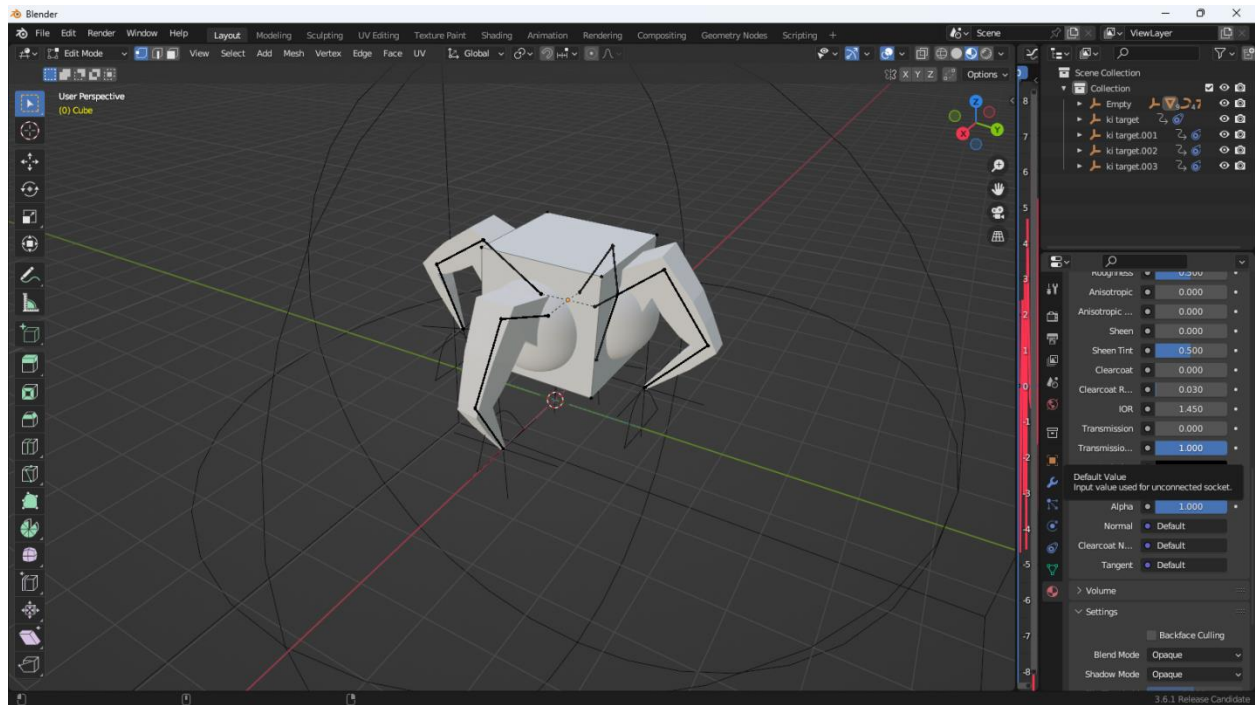
Step 4: Refine the Animation

Move to subsequent frames.

Make changes to the object's properties.

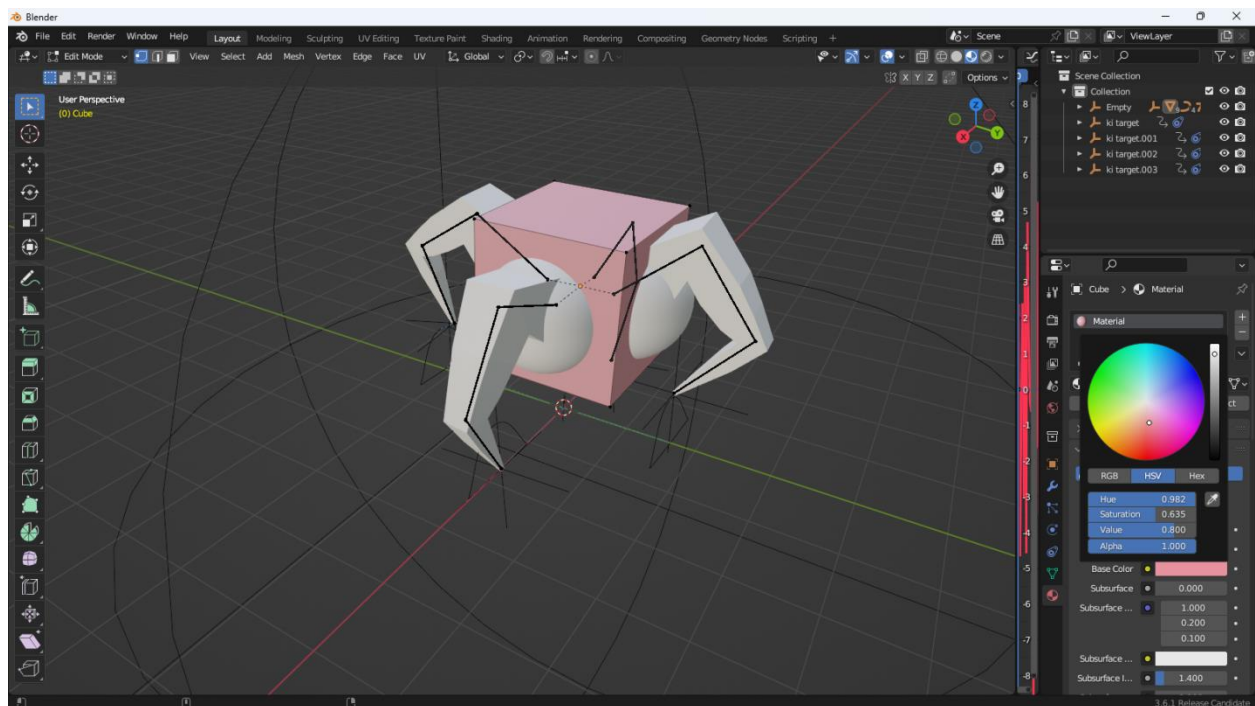
Set keyframes for these changes (press "I" and choose the property).

Blender will interpolate between keyframes to create smooth motion.



Step 5: Add Materials and Textures

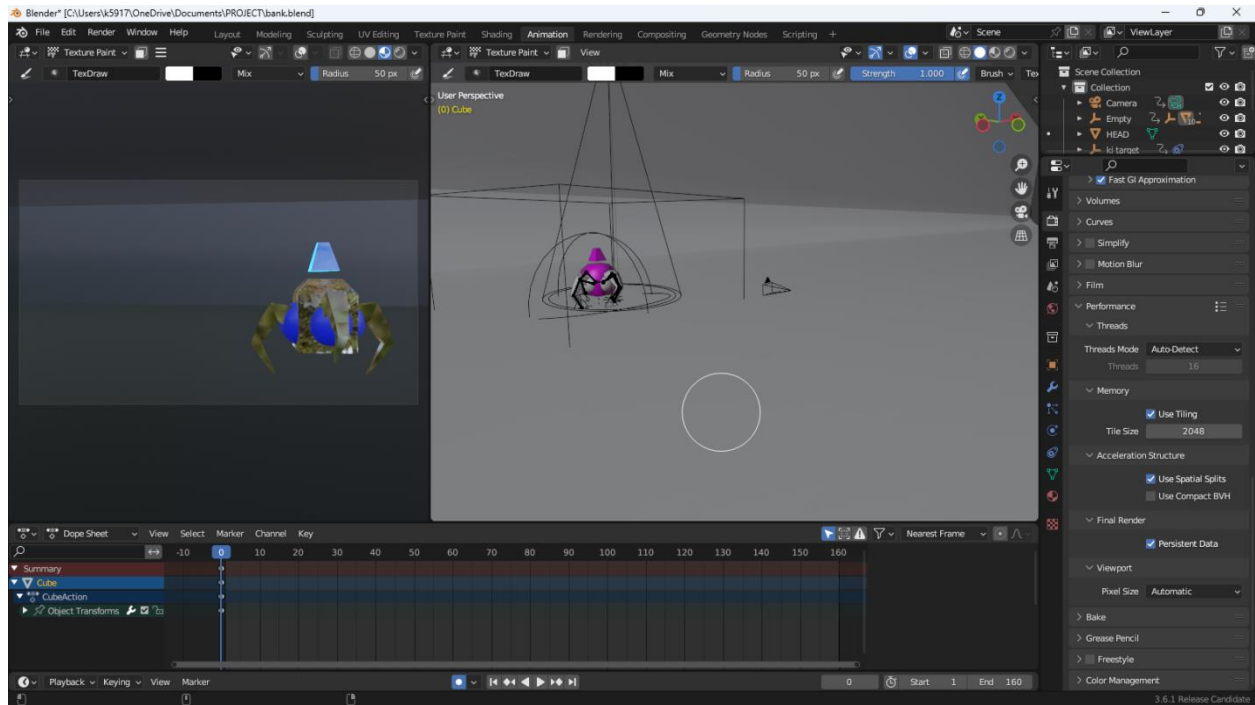
Go to the "Shading" workspace to add materials, textures, and lighting to your objects to make your scene more visually appealing.



Step 6: Set Up the Camera

Position and animate the camera to capture the desired view of your scene.

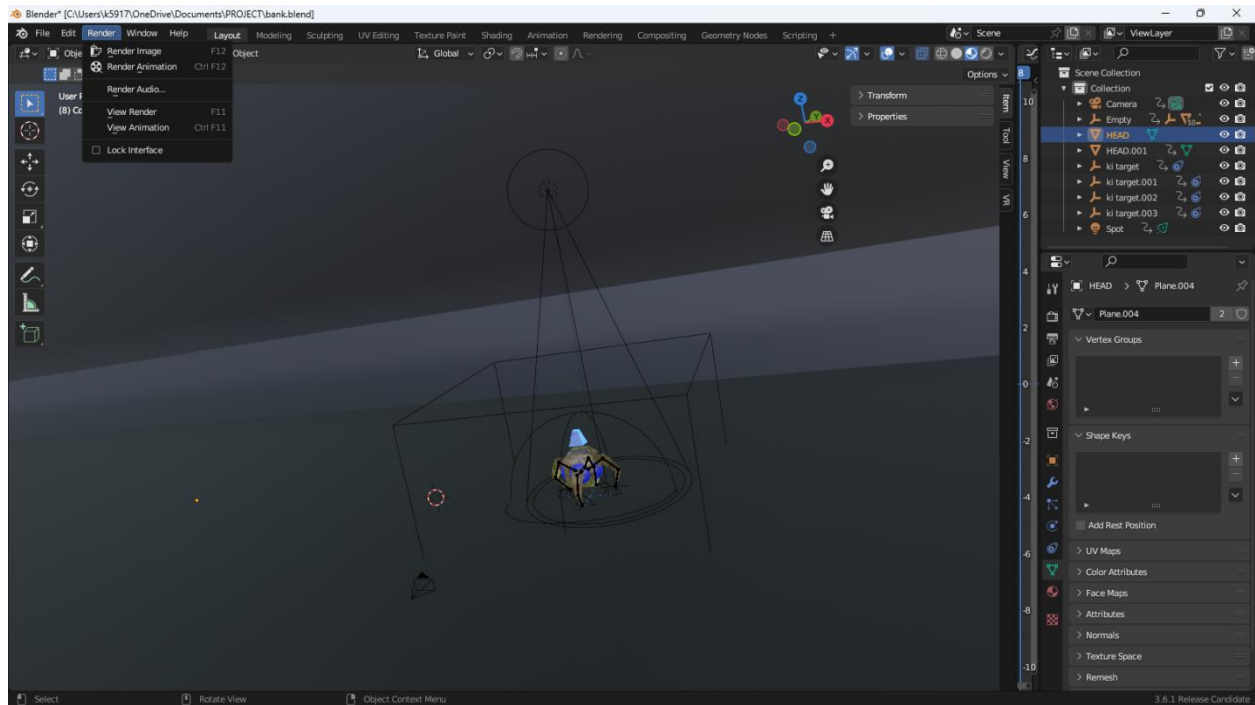
Create keyframes for the camera's location and rotation as you did for objects.



Step 7: Adjust Render Settings

Go to the "Render" settings to configure resolution, frame range, file format, and output location.

Select a render engine (Cycles or Eevee) and set up lighting and render settings accordingly.



Step 8: Render the Animation

Click "Render" to start rendering the animation.

Depending on your scene's complexity, rendering can take a considerable amount of time.

Step 9: Save and Export

Save your Blender project.

Export your animation to a video format (e.g., MP4) or image sequence (e.g., PNG).

OUTPUT:



RESULT:

Thus the 3D animation has created and verified Successfully.

EX.NO : 6	WORKING WITH E-LEARNING AUTHORIZING TOOLS
DATE :	

AIM :

To create simple E-Learning module for a topic of your choice by using E-learning Authoring Tools.

ALGORITHM :

1. Sign Up and Log In:

- Visit the Moovly website (<https://www.moovly.com/>).
- Sign up for an account if you don't already have one.
- Log in to your Moovly account.

2. Dashboard:

- After logging in, you'll be taken to your dashboard.

3. Create a New Project:

- Click on the "Create a Project" button or a similar option on the dashboard.

4. Choose a Template (Optional):

- You can start from scratch, but Moovly offers various templates for e-learning content. Select one that fits your needs or start with a blank canvas.

5. Workspace:

- You'll be taken to the Moovly workspace, where you can start building your e-learning content.

6. Add Objects:

- Use the Moovly library to add various multimedia objects like images, videos, text, shapes, and more to your project. You can upload your own media or use Moovly's built-in resources.

7. Animate Objects:

- Add animations and transitions to objects to make your content interactive and engaging. You can set entry and exit animations, motion paths, and other effects.

8. Narration and Voiceovers:

- You can add voiceovers or narration to your content. Record your own audio directly in Moovly or upload pre-recorded audio files.

9. Text and Captions:

- Use text boxes to add captions, titles, and descriptions to your content. You can customize fonts, styles, and animations for text.

10. Timeline:

- Moovly uses a timeline-based approach for sequencing and timing. Arrange objects and animations on the timeline to control when they appear and how long they stay on screen.

11. Interactivity:

- Create interactive elements like clickable buttons and links to other sections of your e-learning content or external resources.

12. Preview:

- Periodically preview your project to check how it looks and behaves. Make adjustments as needed.
- 13. Publish and Share:**
 - Once your e-learning content is ready, you can publish it in various formats, such as video, HTML, or as a URL. Choose the format that suits your e-learning platform or audience.
- 14. Testing:**
 - Before sharing your content with your learners, test it to ensure that all interactive elements work as expected.
- 15. Sharing:**
 - Share your e-learning content with your target audience. You can provide links or embed the content on your website or learning management system.
- 16. Analytics and Feedback:**
 - Moovly often provides analytics and tracking features to monitor how learners interact with your content. Use this data to make improvements.
- 17. Update and Maintain:**
 - Regularly update your e-learning content based on user feedback and changes in your curriculum.
- 18. Support and Resources:**
 - If you encounter issues or need help, Moovly usually offers support and resources like tutorials and documentation.
- 19. Backup and Storage:**
 - Ensure that you back up your project files and media to prevent data loss.
- 20. Feedback and Evaluation:**
 - Collect feedback from learners and evaluate the effectiveness of your e-learning content. Use this information to make improvements for future projects.

PROCEDURES :

Step 1: Define Your Learning Objectives

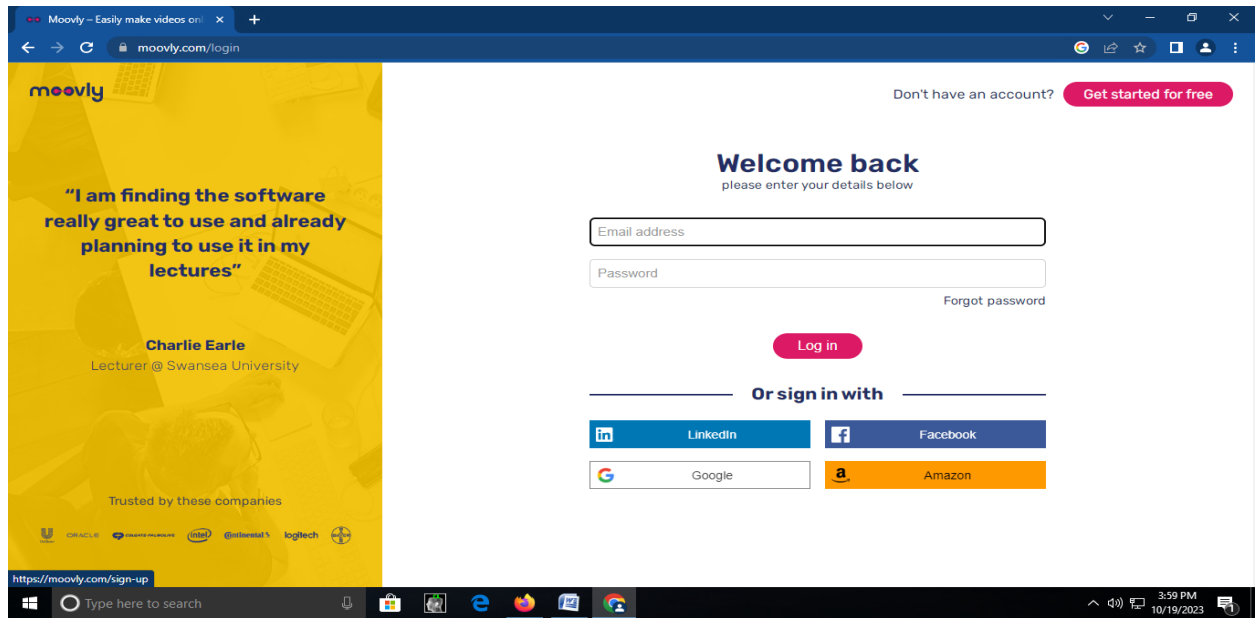
Before you start creating your e-learning module, clearly define the learning objectives. What do you want your learners to know or be able to do after completing the module? This will guide your content creation.

Step 2: Gather Content and Resources

Collect all the content and resources you'll need for your e-learning module. This may include text, images, videos, audio, and any other materials relevant to your topic.

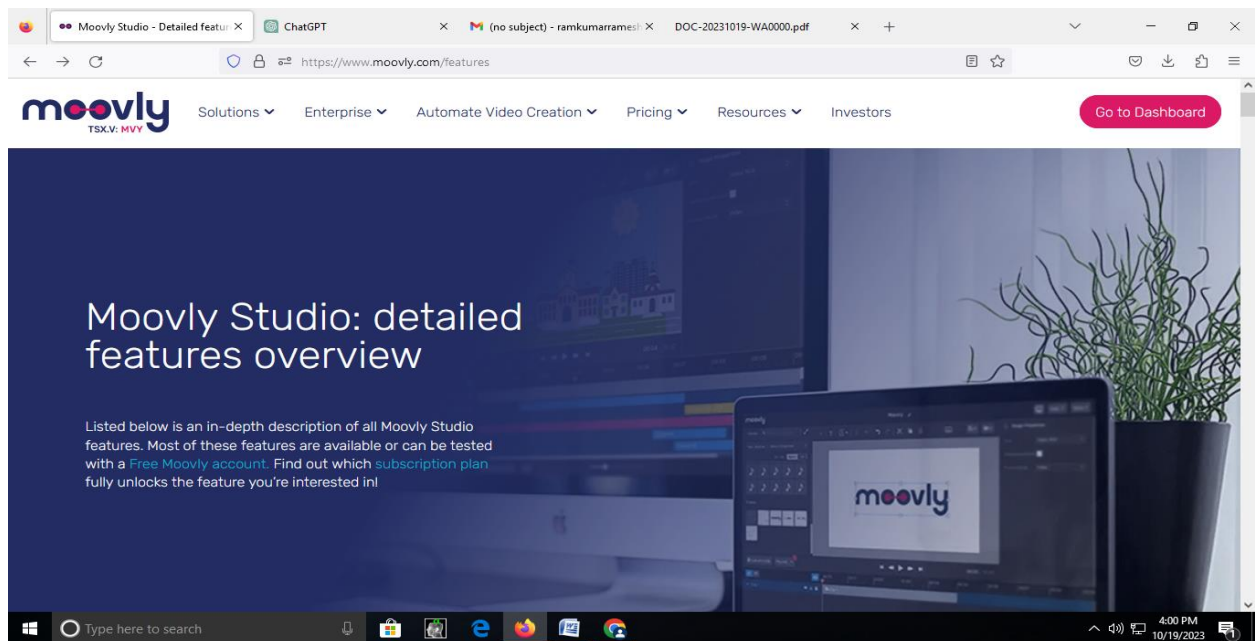
Step 3: Log In to Moovly

If you don't already have a Moovly account, sign up and log in to your account.



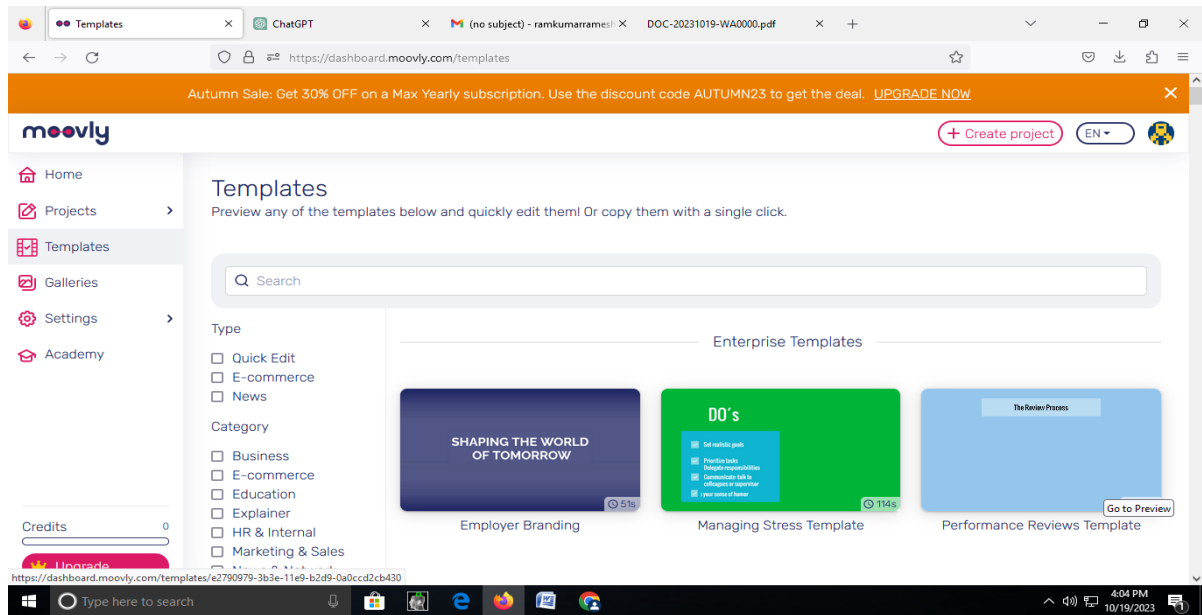
Step 4: Create a New Project

On the Moovly dashboard, click on the "Create a Project" or "New Project" button.



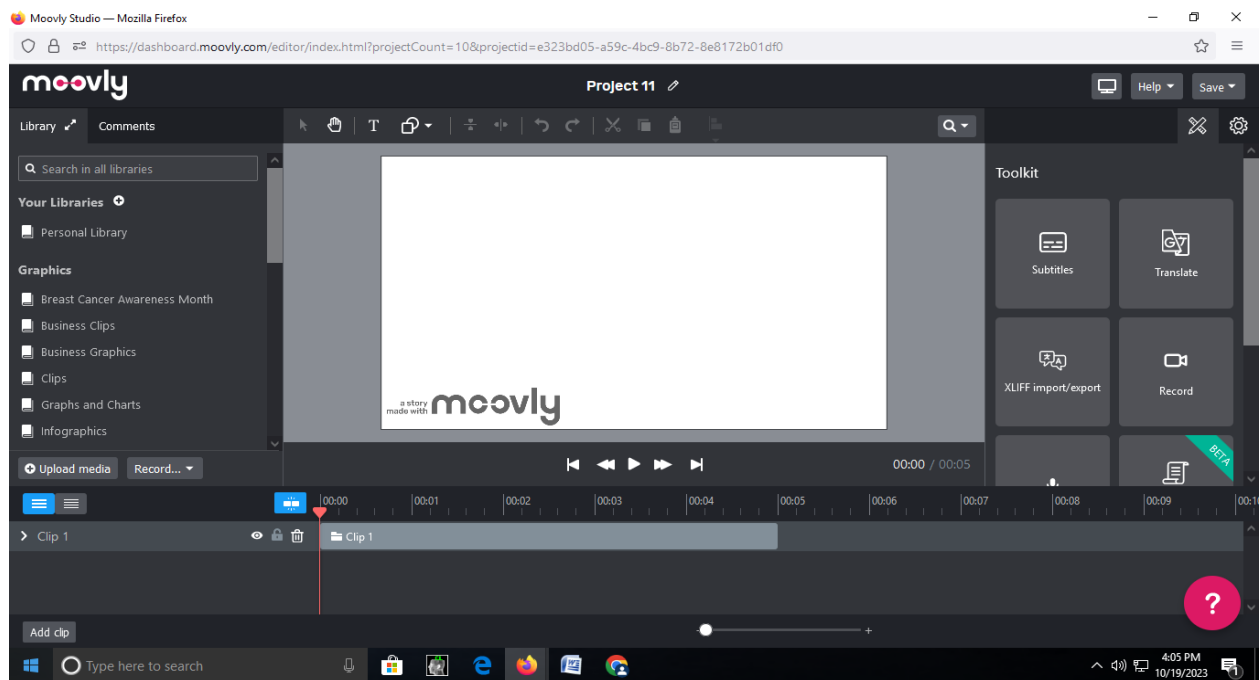
Step 5: Choose a Template or Start from Scratch

You can start from scratch or choose a template that fits the structure and design you have in mind for your e-learning module.



Step 6: Build Your E-Learning Module

Here's how to build your e-learning module within Moovly:



a.

Create Sections:

Divide your module into sections or chapters to organize the content logically.

b. Add Slides:

Within each section, create individual slides. Slides will hold the content for that section.

c. Add Content:

Populate each slide with content. Add text, images, videos, and any other multimedia elements to convey the information.

d. Animate and Sequence:

Use Moovly's animation features to make the content engaging. You can set animations for objects and transitions between slides.

e. Add Narration:

For each slide or section, you can add voiceovers or narration. Record your own audio or upload pre-recorded audio files.

f. Add Interactivity:

Create interactive elements such as buttons or clickable areas that link to other sections or external resources.

Step 7: Preview and Test

Periodically preview your e-learning module to ensure that it looks and functions as intended. Test all interactive elements to ensure they work properly.

Step 8: Publish and Share

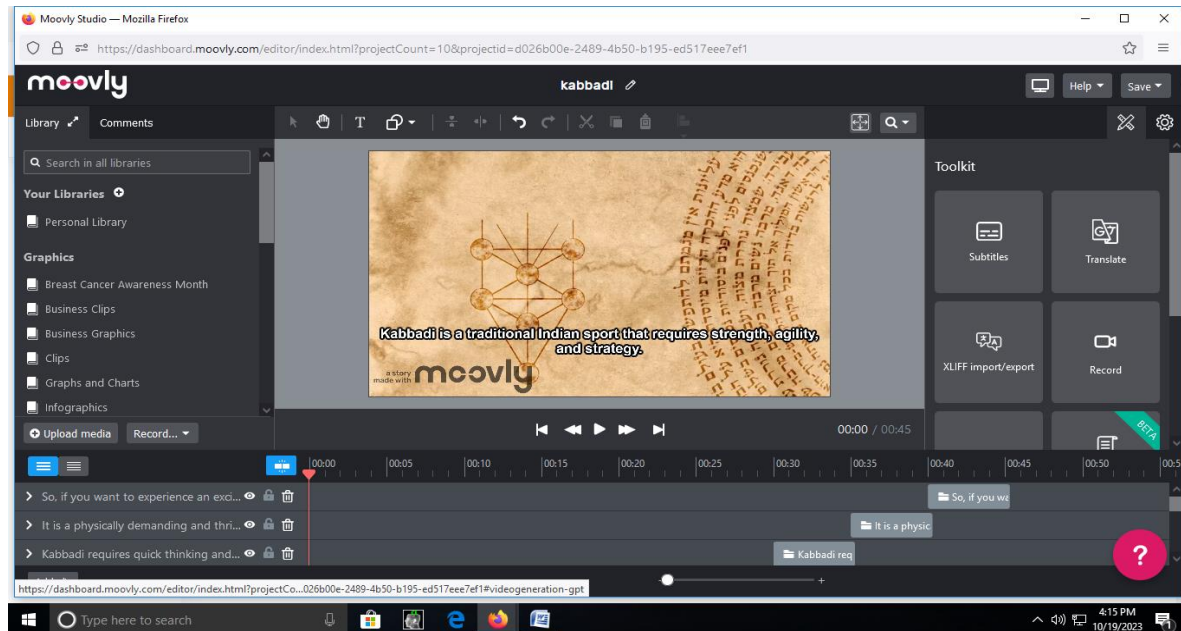
Once you're satisfied with your e-learning module, publish it in a format suitable for your audience. Options typically include video, HTML, or URLs.

Step 9: Share and Gather Feedback

Share your e-learning module with your target audience, whether it's through a learning management system or other means. Collect feedback from learners to make improvements.

Step 10: Evaluate and Update

Continuously evaluate the effectiveness of your e-learning module based on learner feedback and results. Make updates and improvements as needed.



RESULT :

Thus the simple e-learning module using moovly tool was implemented and verified Successfully.

EX.NO : 7	WORKING WITH VR AND AR APPLICATIONS
DATE :	

AIM :

To create simple VR and AR Applications by using ADCore Tools.

ALGORITHM:

Step 1: Install VR Viwer like ARCore.

Step 2: Define Your Project Goals.

Step 3: Choose the Development Platform.

Step 4: Adding movement to the model.

Step 5: Design the User Experience.

Step 6: Gather Necessary Assets.

Step 7: Develop the Application.

Step 8: Testing and Quality Assurance.

Step 9: Optimize for Performance.

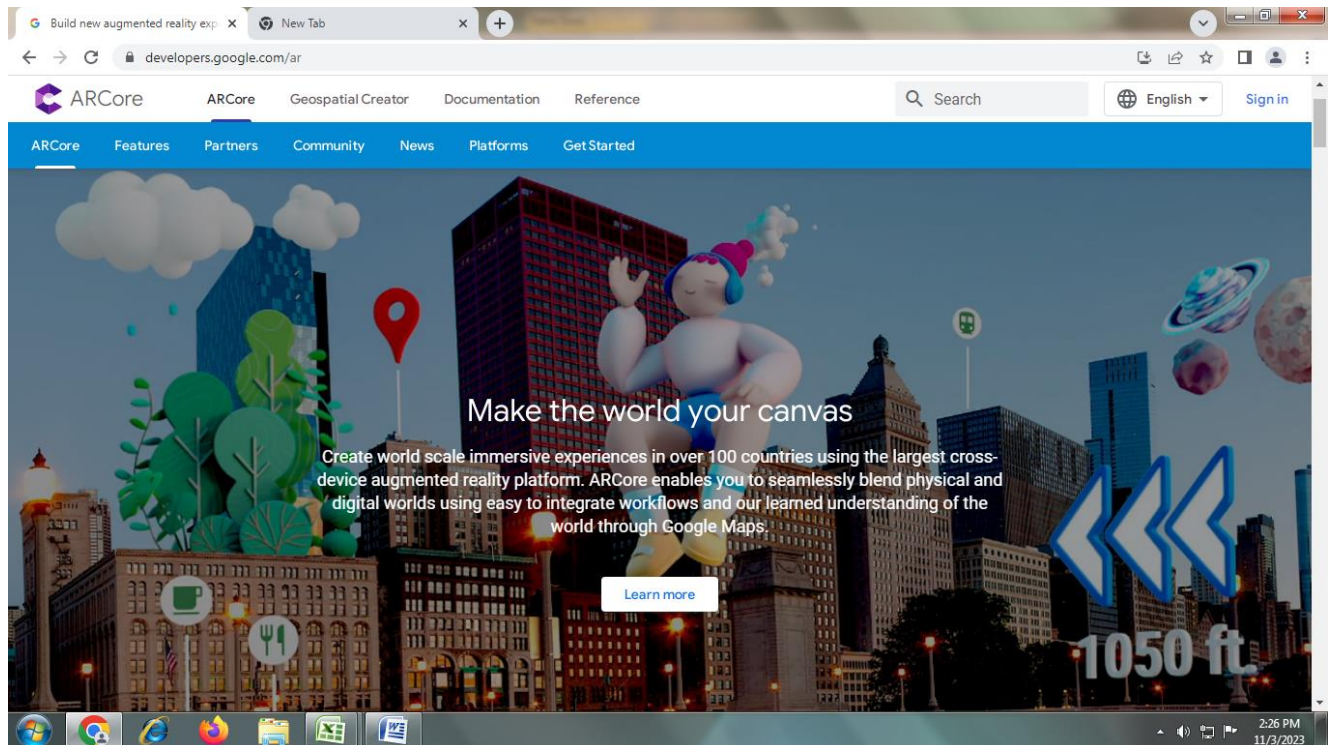
Step 10: User Testing and Feedback.

Step 11: Deployment

PROCEDURES :

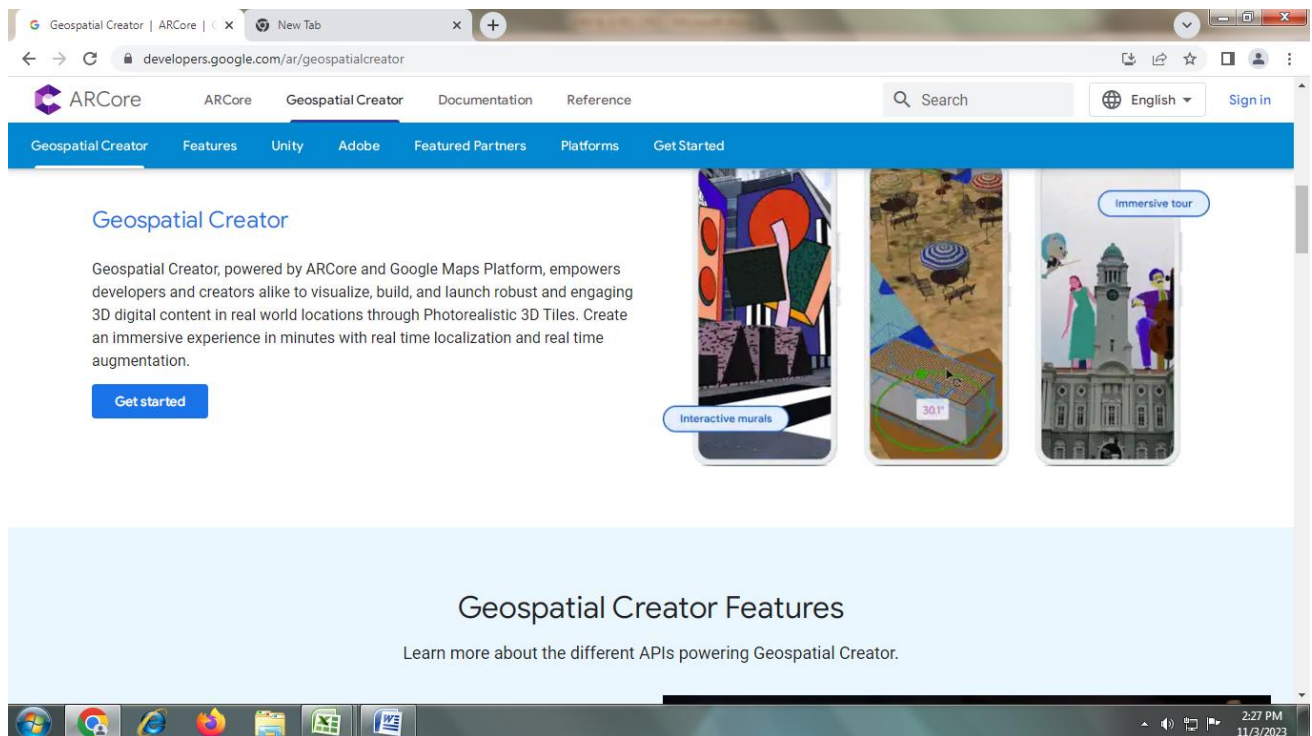
1. Define Your Goal:

Begin by clearly defining the purpose and objectives of your AR application. Understand what problem you want to solve or what kind of experience you want to offer.



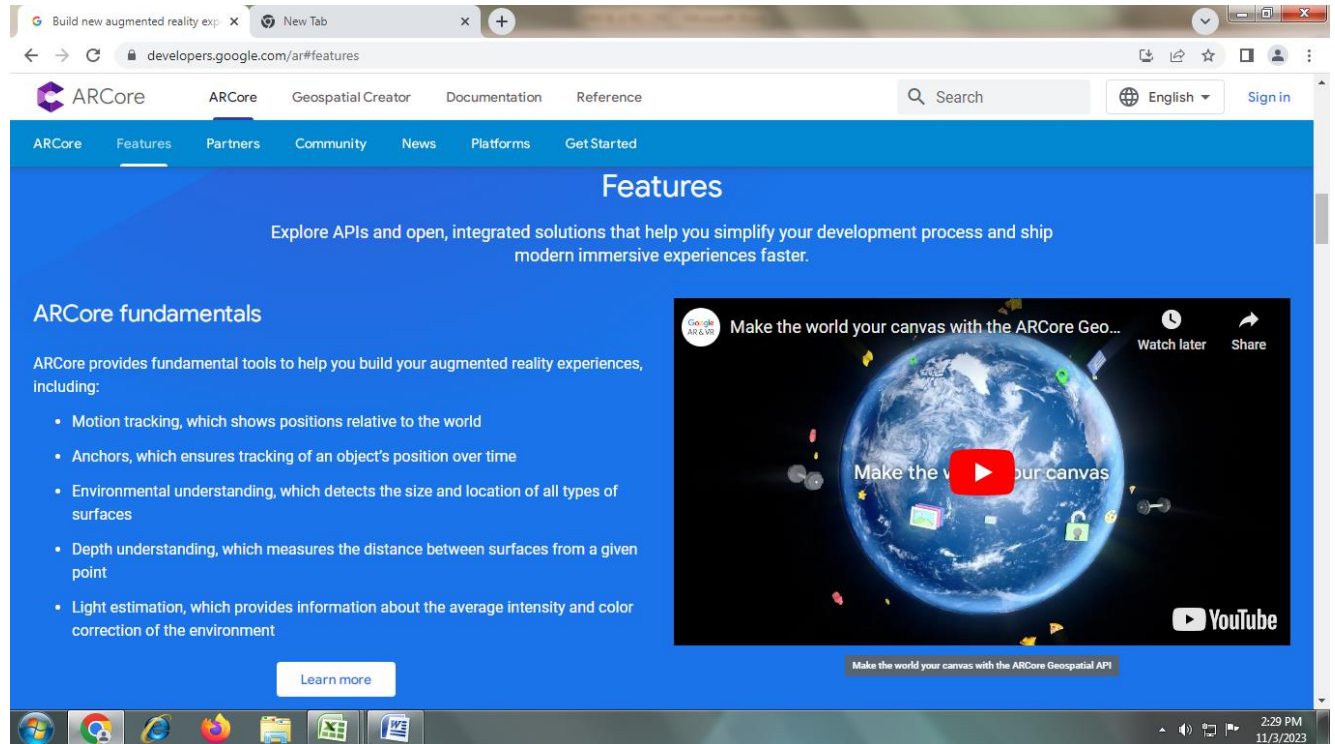
2. Choose AR Technology:

Select the appropriate AR development platform based on your target devices. Common choices include ARKit for iOS, ARCore for Android, or cross-platform frameworks like Unity with AR Foundation.



3. Design User Interface (UI/UX):

Create a detailed design for your AR app's user interface and user experience. This includes designing menus, overlays, and interactive elements.



4. Gather Assets:

Collect or create 3D models, textures, animations, and audio assets required for augmenting the real world effectively.

5. Develop the App:

Write the code for your AR application. This involves implementing features such as object recognition, tracking, and user interactions.

6. Testing

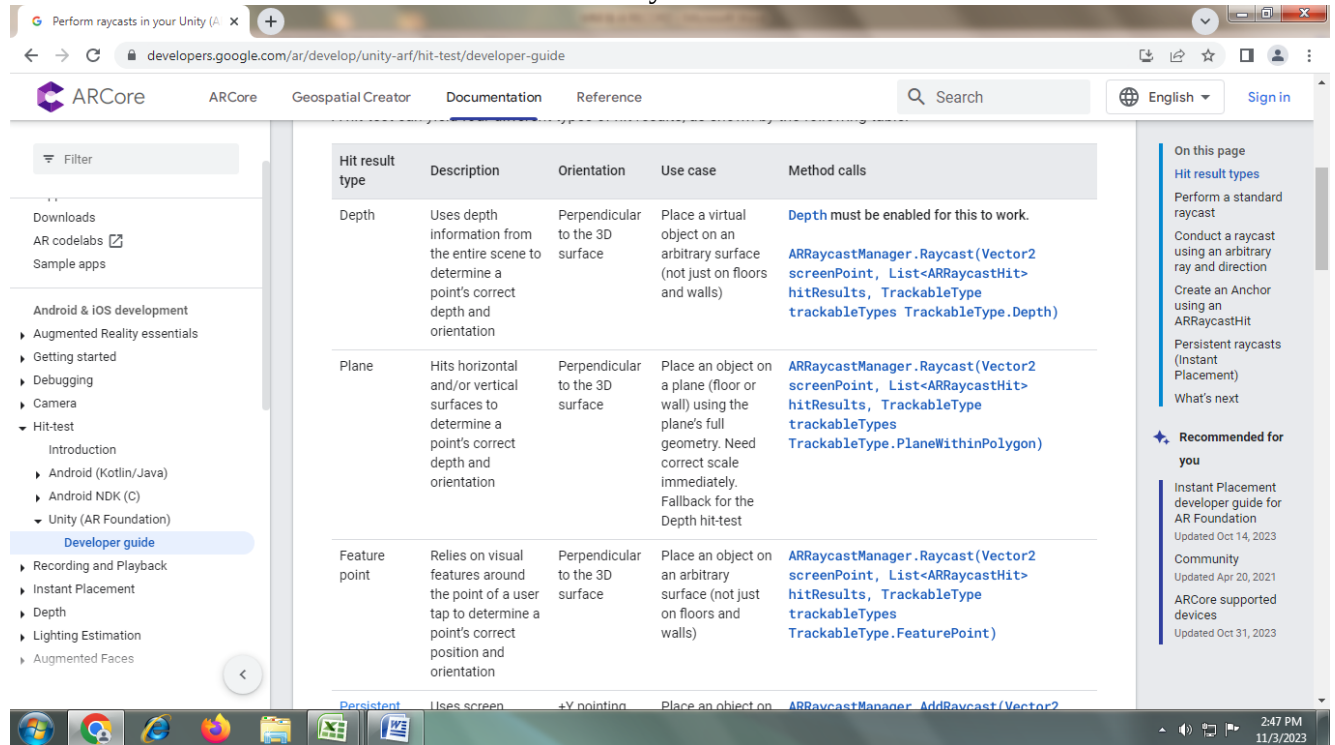
Extensively test your AR app on a variety of devices to ensure that it functions correctly and delivers a smooth user experience.

7. Optimize Performance

Optimize your AR app for performance and battery life, as AR applications can be resource-intensive. This may involve optimizing rendering, reducing memory usage, and conserving power.

8. User Testing

Conduct user testing with a diverse group of users to gather feedback on usability, comfort, and any issues they encounter.



9. Deployment

Once your AR application is ready, publish it on app stores such as the Apple App Store or Google Play. Follow the platform-specific guidelines for app submission.

10. Maintenance

Regularly update and maintain your AR app. This includes fixing any bugs, adding new features, and ensuring compatibility with the latest AR technologies and devices.

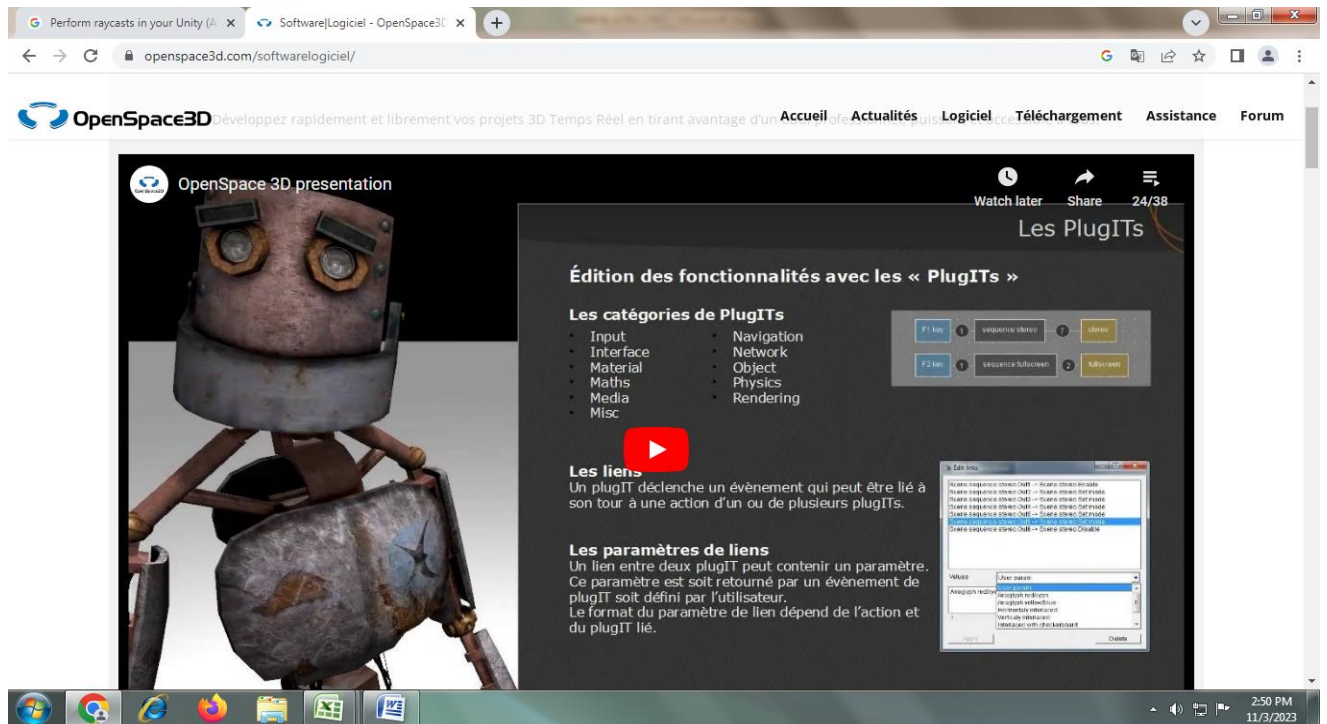
Working with VR Applications:

1. Define Your Goal

Clearly define the purpose and objectives of your VR application. Consider the type of experience you want to create, whether it's a game, simulation, training, or other applications.

2. Select VR Platform:

Choose the VR platform you'll develop for, such as Oculus Rift, HTC Vive, PlayStation VR, or mobile VR platforms like Oculus Quest.



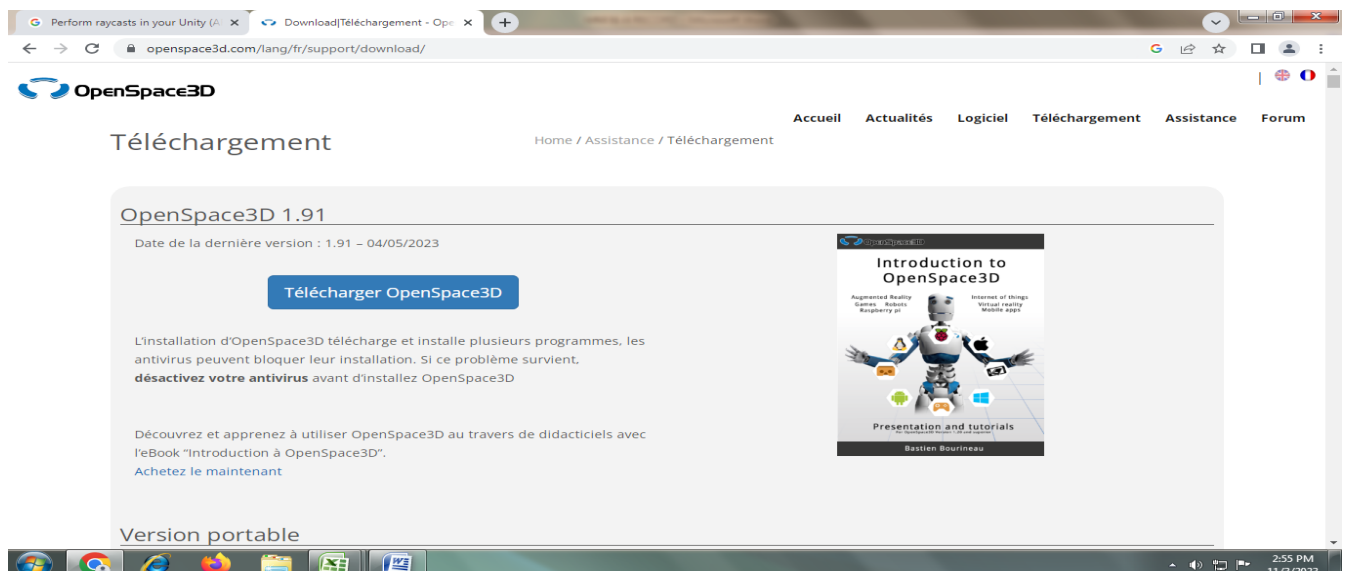
3. Design VR Environment:

Create a detailed design of the 3D virtual environment, considering user immersion, navigation, and interaction.

Plan the layout, aesthetics, and the overall user experience of the VR world.

4. Create 3D Assets

Develop or acquire 3D models, textures, animations, and audio assets for your VR application. These assets are essential for creating the virtual world.



5. Develop the App

Write the code for your VR application, incorporating head tracking, hand controllers, and VR-specific interactions.

6. Testing:

Thoroughly test your VR app on the target VR hardware to ensure it provides a smooth and immersive experience. Pay close attention to performance, frame rates, and comfort.

7. User Testing:

Conduct user testing to gather feedback on the overall experience, including comfort, user interface design, and any motion sickness issues.

8. Optimization:

Optimize your VR app for performance to maintain high frame rates, minimize latency, and prevent motion sickness. This may involve fine-tuning rendering settings and interactions.

9. Deployment:

Publish your VR app on appropriate VR platforms such as Oculus Store, SteamVR, or other VR marketplaces. Follow the guidelines and requirements for each platform.

10. Maintenance:

Continuously update and maintain your VR application to enhance the user experience, address issues, and remain compatible with evolving VR technologies and hardware.

RESULT :

Thus the creating VR and AR applications using ARCore and Openspace 3D tool was implemented and verified Successfully.